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WAVEFORM 2001

Australasian Computer Music Association Conference

PAPERS

How the computer assists composers: a survey of contemporary practice

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With the increased prevalence of the computer as a compositional tool comes an expansion of the musical genres that constitute "computer music". Traditionally computer music composition was a marginal practice, dominated by a desire to use the computer to create novel music by freeing the composer from the limitations of traditional mediums. As we enter a new millennium the use of the computer for music making is common place and therefore the need to consider the impact of the computer on music composition is greater than ever. This paper will examine the ways in which the computer assists the composer and in doing so will shed light on the complex relationship between the composer and computer, and indicate trends in computer assisted compositional practice. In particular, I will suggest that the computer assists the composer in three ways: (i) by defining the range of compositional choices available, (ii) by acting as a sounding board that reflects back the composer's ideas, and (iii) by being a vehicle that enables composers to realise their musical ideas.

Introduction

The computer is currently the dominant tool assisting composers in their compositional practice, and the diversity of software and hardware configurations means that the ways in which computers support the compositional process are numerous. Computer music systems, often incorporating MIDI keyboards, have almost completely replaced the piano and manuscript paper in the composition studio and computer-based recording systems now dominate recording studios. The shift to computer assisted composition (CAC) for some composers has not changed their traditional compositional practices and processes; however, for most composers the use of the computer has expanded their compositional horizons. This paper will examine the ways in which the computer assists the composer and in doing so will shed light on the complex relationship between the composer and computer, and indicate trends in computer assisted compositional practice. In particular I will suggest that the computer assists the composer in three ways, (i) by defining the range of compositional choices available, (ii) by acting as a sounding board that reflects back the composer's ideas, and (iii) by being a vehicle that enables composers to realise their musical ideas.

With the increased prevalence of the computer as a compositional tool comes an expansion of the musical genres that constitute "computer music." Traditionally computer music composition was a marginal practice, dominated by a desire to use the computer to create novel music by freeing the composer from the limitations of traditional mediums. This experimental, or *avant- garde*, attitude is consistent with the early adoption of many new music technologies, for example, the adoption of analogue electronics and tape recording which lead to the development of Elektronische Musik and Musique Concrète respectively. Early computer music making consolidated predominantly in what is called electroacoustic music.

Computer music composition in the early part of the 21st century can no longer be so specific that it defines itself by one style. While acknowledging the historical connection between CAC and electroacoustic music, the computer as a musical tool needs now to be considered genre-neutral. The computer, and more specifically the process of digitisation, has been a significant influence in the dissolution of boundaries between various musical genres in the latter half of the 20th century. In this

paper I will focus on the current state of CAC and defer debate regarding the influence of the computer on the post-modern condition (Lyotard, J.-F. 1984) to another forum. Therefore, in this paper, comments about CAC will take into account the rich diversity of musical genres in which the computer now plays a significant role, including electroacoustic, instrumental, choral, electronica, rock, and world music.

The findings in this paper are particularly influenced by my case study analysis of five composers during the late 1990s; David Hirschfelder, Steve Reich, Brigitte Robindoré, Paul Lansky, and David Cope. Examples from the practices of these composers will be used as examples.

The ways in which computers assist composers will be discussed under three broad headings, *increasing choice*, *simulation*, and *enabling*. After this I will discuss some aspects of the collaborative partnership that develops between composers and the computer, and conclude with comments about future directions of CAC systems.

Increasing choice

According to Mashall McLuhan (McLuhan, M. 1964) the first stage of acceptance of a new technology involves simulating previous media, which facilitates the smooth integration of the technology into existing practices. This appears to be correct in relation to the widespread acceptance of a technology, for example in activities such as writing, accounting, telephony and so on. It is also a reasonable description of the use of computers in supporting established compositional processes, such as the automation of score publishing. However, it does not adequately explain the introduction of new technologies, including the computer, to creative activities such as composition. The creative use of technologies seems to include its misuse as much as its use.

By definition, composers (and other creative people) are looking to exploit new techniques and offer new perspectives on existing material and so, to them, technologies provide opportunities to explore new avenues of musical expression. The computer, especially in its ability to change characteristics through re-programming, assists the composer by providing a rich source of new possibilities. The compositional appetite for new software programs, new sounds and synthesis processes, and accesses to more manipulative features seems insatiable.

The computer seems most competent at meeting this demand and is able to provide choices in many aspects of the compositional process. These aspects include the nature of capturing musical ideas and material, their representation and communication, the modification and processes available for their development, and the ability to control aspects of interpretation and performance (to produce not simply invent).

Contemporary computer music systems predominantly capture musical ideas as gestures, either via MIDI or audio capture. The distinction between MIDI sequencing software and hard disk recording systems continues to blur and this reflects their underlying similarity as systems that focus on gesture capture and modification. Such systems continue the heritage of composition with recording devices begun by Pierre Schaeffer, Pierre Henry and others.

In addition to recording gestures, the computer can allow composers to describe their musical ideas in a great variety of ways. Predominant amongst these are the use of common practice notation on a five line stave (CPN), step sequencing often using a series or matrix organisation, visual programming environments that extend the heritage of *Music N* systems and which depict musical processes as flow charts, and direct alpha-numeric linguistic descriptions most commonly using a general purpose computer programming or scripting language. This wide variety of descriptive choice is reflected in the diversity of software in this category, including *Finale* and *Sibelius* for notation, *Rebirth* and *M* for step sequencing, *MAX* and *AudioMulch* for visual description of process, and *Csound* and *jMusic* for text-based description.

While the methods of describing the music to the computer are quite varied, the processes of modification are fundamentally similar. They are similar in that the musical representation in the computer is a digital one and the transformation of captured, or described, musical source material is always conducted in this medium. There is, however, one broad differentiation between digital music representations that can be made, that between music as events and music as sound. Event representation (such as in MIDI messages, step sequencers, or *jMusic* scores) are more susceptible to traditional compositional treatments focussing on pitch and rhythm variations, while audio representations (as used in samplers, signal processing systems, or hard disk recorders) lend themselves most easily to timbral and time based modifications.

In the area of event structuring the computer offers the composer few "new" treatments but, because of its speed and programmability, can significantly increase the efficiency of many compositional processes. For the composer this can mean faster compositional turnover, the ability to experiment with a greater number of options, or the ability to perform operations previously too difficult.

The computer is particularly influential for composition today in its ability to handle audio data. From digital remastering of analogue orchestral recordings, to sample and synthesis collages assembled in digital multitrack systems, and to music distribution in compressed formats such as MP3, it seems that the computer's ability to manipulate digital sound is all pervasive. The most significant aspect of this for composers is the ability to produce a final audio product more than the types of manipulations that are possible. This blurring of the lines between composer and performer is well established for those in electroacoustic music circles, as is fundamental to the emergence of the computer in rock, pop, and electronica. The computer blurs traditional boundaries between writing tool and performance or presentation instrument, and in the process forces a redefinition of the term composer that includes the role as producer and distributor.

With the continuing emergence of the internet as a mode of music distribution, in particular the rise of peer to peer systems such as *Napster* and *Gnutella*, the computer assists composers in communicating with their audiences quite directly. It is inevitable that peer to peer systems will have an impact on collaborative compositional efforts as well. Paul Lansky is one composer who considers the digital distribution of music at least as important as the digital creation of the music, he writes that, "Before the advent of recording the only way one could be a sound-giver was to be a performer. Today, however, most of us would have to admit that giving and receiving sounds in one way or another is the most active part of our musical social life" (Lansky, P. 1990).

The final choice I wish to address, provided by the computer for the composer, is the option for composers to involve themselves in computer music system design and instrument creation. It is not that such activities were not always of interest to composers, for example J. S. Bach is reported to have had quite a career in design and maintenance of pipe organs, but that the skills of composing and instrument making on the computer can blur quite easily. In particular this is evident for composers who compose in computer programming environments. The skills required to communicate musical ideas and to build new musical tools can be one and the same. In particular the abstracting tendency of algorithmic composition can easily lead to abstractions or structure or design that can be applied beyond the immediate situation—effectively creating a generalised tool. This blurring of activities is even more accessible to composers through visual programming environments such as *AudioMulch* and *MAX*, and is accessible to some degree in sequencing software including *Logic* and *Cubase*. Virtual synthesizer technologies such as Reason enable construction of new "patches" via virtual patch-leads to create new synthesis configurations and instruments. For some composers the opportunity to create new composition tools and instruments becomes a significant, even dominant, part of their musical life. For example, David Cope is well known more as the developer of the *Experiments in Musical Intelligence* (EMI) software that replicates musical style, than for his own compositions. This is ironic given that Cope's original motivation for *EMI* development was to assist his own compositional processes.

The computer assists the composer by providing new and expanding choices about the type and extent of music constructing activities, and by being central to many activities related to music creation into which the composer can choose to extend themselves.

Simulation

Because music is a temporal art and sound is fleetingly present the ability of the computer to "hold" the music greatly assists the composer. Paper scores and tape recordings provide similar facilities but the particular advantage of the computer to that it can hold both score and sound, and translate between them. This ability is particularly obvious in a system which combines an MIDI sequencer with software synthesizer, where the symbolic score, held as MIDI data, can be transposed, elongated and so on, while the timbre of each event can be controlled by adjusting the synthesis parameters or waveform source material. The computer goes beyond mere storage of the music by being programmed to manipulate the stored data or to create new data, thus enabling the audition of a variety of simulations and interpretations of the piece.

Compositional technologies have long been used because of their ability to reflect or articulate the composers ideas. An acoustic instrument, for example, is useful for sounding out a musical theme, testing harmonic relationships, or auditioning rhythmic ideas. In so doing acoustic instruments both amplify and interpret the musician's gestural expressions. Such feedback and amplification is a valued feature of many computer music systems which, like acoustic tools, provide real time feedback that enables a interactive and intuitive compositional process to proceed. However, feedback need not be real time to be effective. Some computer systems, particularly programmable ones, require time to compile and or render before providing feedback. Programs such as *Csound* have generally operated in this way but still provided effective compositional assistance. Audio rendering of event based scores is a translation process used by computer musicians. These processes have long been used for elecroacoustic music with software such as *Csound*, *Cmix* and more recently *Supercollider* and *jMusic*.

The provision of real time audio feedback is becoming an question of degree these days with computer processing speeds well able to support real time audio processes. The issues for the composer now relate to quality and quantity of the feedback they desire, and the degree of manipulation they can comprehend. The cognitive load of the creative process comes into play with regard to the number and complexity of real-time adjustments a composer can usefully consider at one time.

Computer systems additionally provide visual feedback in the form of scores, waveforms, flow charts, statistics, and other representations of the music. Through audible and visual presentation of music, computers assist the composer to hear and see their work from many perspectives, to simulate ideas, and to experiment with alternate variations.

Capture and feedback are process of communication between the composer and the computer that involve interpretation. The composer translates ideas into gestures or symbols which are stored in the computer. The storage and re-presentation of those ideas requires the computer to express them as best it can. A simple example of the limitation of the computer translation is the quantisation by MIDI systems of pitch and dynamic values into 128 steps, or the quantisation of audio systems of high frequency sounds using just a few sample values per cycle. Despite the ease of pointing to limitations, in many cases the current resolution of digital audio data is adequate and the process of translation is transparent. Another area in which the computer assists in translation is between symbol systems, for example in the translation from MIDI gesture capture to CPN in programs such as *Sibelius* or *Finale*. In select cases, such as CPN scoring, the translation may be preferable to the inevitable inconsistency of human expression. Other examples of the interpretation being preferred include the quantisation of performed rhythms in pulse-based music, or the dynamic compression inherent in MP3 encoding which, for some, improves the presence of their music.

For humans, the role of interpretation of musical intent is highly valued, but generally in machines it is not desirable—often because the machine interpretation is insensitive. However, in electronica the characteristic interpretation of some machines has become part of style, for example the regular rhythmic subdivision and the "sound" of particular analogue drum machines and synthesizers. As the computer became more a part of this community there were attempts to mimic the characteristic "sound" of particular machines through sampling or virtual analogue modelling. By simulating other machines, the

computer assists composers to achieve their result without recourse to the original methods or mediums. Similarly, attempts to simulation human interpretive nuance have been used in the areas of score performance, typographical design in score publishing programs, and accompaniment of live performance in well established musical styles.

Another aspect of the computer's symbolic processing ability is that it can be applied to musical analysis and various pattern recognition tasks. Software systems, such as *CUE*, *PatchWork* and *jMusic*, include musical analysis capabilities and these and other programmable systems enable the composer to create their own analysis routines. David Cope, for example, uses the music analysis function in *CUE* to provide statistical feedback while composing (Cope, D. 1997). He finds it useful to check aspects such as the pitch class distribution, common harmonic relationships, and so on, to add to his intuitive sense of the compositional development.

Computer analysis of audio data can be used to provide information such as beat and pitch tracking or the spread of harmonic spectra. The application of these techniques is evident in programs such as *Logic* and *Cubase*, where they are used to translate monophonic audio recordings into CPN scores, and to adjust the timing of MIDI tracks to synchronise with recordings of live performances on audio tracks. The ability of the computer to analyse music in these automated ways extends its ability as an assistant beyond the static nature of many previous music technologies. The interactive nature of the feedback through analysis takes the notion of the computer as translation device to a new level.

As with all translation, something is altered in the process, and to the extent that it saves time or improves results it is welcome, and to the extent that it distorts or inhibits clear musical expression it is unwelcome. In any case, translation is a feature of the computer in its functioning as a sounding board for the composer. The computer system as sounding board will resonate in a particular way, accentuating some aspects of what is communicated to it and dismissing others.

Enabling

In many ways the contemporary computer music systems are like modern four wheel drive recreational vehicles, most of which are owned by city dwellers who go off road on the occasional weekend. The computer music system is a vehicle with which the composer explores musical space, usually staying on the known roads and following the paths of least resistance provided by the system's design. Such usage may well indicate an effective computer music system, one that is generally efficient for creating the music it was designed to assist with, and able to move into uncharted territory when required. Most MIDI/audio sequencing/recording software fit this description for those writing music in the most popular genres. It is interesting to consider what sort of vehicle your computer music system similar to? Is it a hot rod, a bicycle, a tank, or a family-sized station wagon?

As a vehicle for exploring sonic space or musical territory, the computer music system is often expected to go "where no man has gone before" in the search for novelty and the uncovering of new methods of expression. In most cases, however, new ideas come from the composer and the computer assists as an enabler of those ideas rather than encouraging them. In this relationship the importance of clear communication between composer and computer, as discussed above, is critical. The effectiveness of the computer system from this perspective lies in its ability to help realise the compositional intentions of the composer. This is made additionally awkward because those intentions are rarely clear in the mind of the composer, usually evolving and clarifying as the compositional processes proceeds. The computer music system can act as an enabling device bridging the gap between conception and realisation as sound or score.

In helping realise the composition, the computer system typically works in the symbolic domain, audio domain, or both. Exploration of the sonic space has dominated computer music activity because of the computer's particular ability to work with sound through sampling and synthesis. This ability has allowed the computer to imitate other sounding devices—including acoustic instruments, the human voice, and sounds of other machines—as McLuhan's theories (refereed to above) predict, as well as to create new

sounds. Significant work in the computer music community has resulted in the development of synthesis and signal processing techniques that have opened up vast new vistas of sonic territory for exploration. The balance of imitative and innovative audio treatments in computer assisted composition is particularly evident in modern film scores which typically include music in a range of genres, much of which is produced with the one computer system. For example, David Hirschfelder's film scores often include orchestral works which combine sampled sounds and live recordings, as well as sections of synthetic sound scapes, and a popular song that typically consists of live vocals and guitar over a computer generated backing. All of these are created in the one CAC environment.

In the symbolic domain, the computer system generally provides a temporal canvas upon which the composer arranges the musical symbols into a structure; CPN scoring software typically works in this way. For the composer, the metaphorical similarity makes the transition from paper score to computer screen quite easy. The "arranging" pages on many sequencing programs provide a similar canvas metaphor on which the composer manually organises the musical objects. Other computer music systems, not surprisingly those that require some programming expertise, go beyond manual editing to enable automated symbolic manipulation. Programs such as *Open Music*, *Common Music*, *AC Toolbox*, *MAX*, and *jMusic* provide composer-definable symbolic manipulations at the musical-event level. By using this capacity composers are able to implement known or invented theories and techniques. These can be as simple as canonic or phasing structures to implementations of the Generative Theory of Tonal Music (Lerdahl, F. and Jackendoff, R. 1983) or the sonification of mathematical models of physical systems (Monro, G. and Pressing, J. 1998). The use of the computer to enable the implementation of theoretical concepts was seen in the earliest computer music (Hiller, L. A. and Isaacson, L. M. 1959) and is particularly evident in the music of Iannis Xankis who used the computer to realise many formalised processes, in particular various probabilistic functions (Serra, M.-H. 1993). It can also be seen in the use of cellular automata processes for pitch and rhythms in the software *CAMUS 3D* (McAlpine, K., Mirando, E. and Hoggar, S. 1999), and for timbral changes through filtering (Kreger, T. 1999).

As well as enabling the creation and production of music, the computer can assist the composer in its distribution. The burning of music onto CDs and the compression of audio for internet distribution as *MP3* files, or as streamed formats including *RealAudio* and *ShockWave/Flash*, has become commonplace. With distribution assistance on hand the composer is able to use the computer at all stages of the music making process, from conception to audience delivery.

The computer music system is able to assist the composer at all stages of the music making journey. At each stage the computer assists to the degree that it enables composers to realise their intentions, to explore pathways, and to help reveal new sonic possibilities. It assists in capturing and solidifying those ideas and sounds into a completed musical form.

Partnership

For the most part this paper has proceeded as a homage to the abilities of the computer music system to assist the composer. My main objective has been to survey the ways in which contemporary computer music systems act to assist the composer, and I have written elsewhere about the significance of the composer in realising this potential assistance through engagement with the compositional process. (Brown, A. 2000) However, it should be emphasised that creating with tools is a partnership and the degree to which to computer can assist the composer depends upon the fit between the two.

The musical thinking of the composer needs to align with the design and working methods of the system for there to be an effective partnership. Given that a computer music system generally reflects the values of its designers, alignment between system and composer usually occurs when designers and composers share similar musical cultures and values.

Computer music systems tend to solidify the musical preferences of their time, either for particular music structures, processes, sounds, or other elements of style and aesthetic. This means that composers of a certain age tend to use tools created around the time their musical preferences were forming, and once

they are familiar with those tools their partnership tends to persist. Computer music systems, while evolving (some dramatically) do have a finite life span that is most closely related to the pace of cultural change but is also at the mercy of technological changes and fashions. For example, new music systems today are more likely to be written in *C++* or *Java* than *LISP* or *Objective C*. This is not because of the languages' inability to express musical ideas; languages are favoured because they are currently popular and widely supported.

Computer music systems also tend to focus on a particular genre of music and the specific requirements of that style. While this trend is quite apparent in software it also affects hardware. For example, the use of MIDI keyboards as input devices was once dominant, but a growing number of composers writing electronica prefer to enter music in a matrix-like display than to perform it on a keyboard.

In using computer music systems to automate and make efficient our compositional procedures we are, in the language of Bruno Latour (Latour, B. 1995), *delegating* responsibility for a portion of our work. Computer assisted composition is not simply human-directed composition because the computer does play a role—by assisting it has an influence. However, the more responsibility we pass to the computer music system the greater the expectations on the composer using the system. This might seem counter-intuitive but put more simply, greater emphasis on the unintelligent (even perhaps unskilled) computer requires more of the (presumably) intelligent composer. A misunderstanding of this position is at the root of Microsoft's disempowering useability policy, which assumes that the more responsibility delegated to "wizards" the "easier" the system is to use. On the contrary, the result is a more limited the system and a demand for more ingenuity from users to recover from errors and to get their non-conformist work done. (Microsoft achieves its intention, to make users increasingly dependent on Microsoft software). In Latour's terms, as the computer assumes responsibility for tasks, in its necessarily automated way, the more it relies upon a particular behavioural pattern in order to act correctly—it increasingly *prescribes* what the user must do. The issue of delegation in creative tasks is even more critical than in office work due to the deliberately novel and unexpected implications of creative processes.

The efficiency gained by enshrining practices within computer music systems is the same force that prevents the system from assisting the composer to move beyond those practices. This "two edged sword" of technology is well understood but is, nevertheless, a significant issue in considering the ways the computer assists composition. A system feature that is considered a positive assistance by one composer may well be considered a hindrance by another. For example, the normal process in a MIDI sequencer or music publishing program for all tracks, or parts, to playback at the same tempo is coherent with traditional ensemble performance practice and generally considered a positive feature. However for Steve Reich, who likes to write music with parts running at differing speeds (phasing), such a characteristic would most likely be a frustration. More importantly, for a composer working with that system, the rigidity of that feature may mean they never consider the possibility of music that phases—surely a pity. Computer music systems do assist composers, but they also direct them.

Conclusion

As the computer, as a technology, is increasingly pervasive in our community, the impact of computer assisted composition affects a broader range of musicians and musical genres. No longer is computer music limited to electroacoustic music, but the knowledge of and experience with computer assisted composition in the electroacoustic music community is a valuable resource that can be shared.

A broader definition of the computer assisted composer is also called for more and more music making activities are centred around the digital medium. The computer musician can now be composer, performer, producer, instrument maker, and distributor. The traditionally solitary working patterns of the composer may continue but with the expanded definition of computer musicians the more collaborative nature of the other roles will effect the degree to which the "composer" interacts with other artists and the audience. In particular the expansion of peer-to-peer networking will mean that the computer will provide the composer with even greater collaborative and communicative opportunities.

The computer as a digital device brings its own idiosyncrasies to the compositional partnership. In particular the representational quantisation and the interoperability of various input/output forms play a role in the computer's musical interpretation and presentation of its musical understanding. The exploitation of these in more and more musical genres demonstrates the increasing acceptance of and reliance upon the computer as musical tool and instrument. Electroacoustic musicans, for example, exploit the digitalness through granular synthesis, electronica artists value quantised timing and synthetic timbres, and popular music makes increasing use of digital processing treatments on vocals.

The partnership between computer and composer works best when they share similar musical values. The computer system inherits values from its designers and the context within which it was developed (time, place, and musical culture). The composer chooses systems whose features resonate with their requirements and preferences, and the extent to which this partnership persists depends upon the degree to which each evolves at a similar pace. Computer music systems rely on both their designers and users to evolve new processes and refine others; sometimes composers are very reliant on the system designers and other times they can make changes themselves. Composers therefore relate to the system designers as well as to the system.

The partnership involves a delegation of tasks and responsibilities from the composer to the computer music system. In the process the computer is able to assist the composer, but also to direct them. The delegation often leads to the establishment of habitual patterns of usage that result in characteristic features in the music composed. The computer assists the composer by increasing or directing the compositional choices, but simulating the musical process and possibilities, and by enabling the realisation and communication of compositional ideas and the distribution of resulting outcomes.

The computer has become an invaluable assistant for the composer, one that provides support and direction at each stage of the musical journey from idea, through development and realisation, to distribution. The challenge for the composer to is to take advantage of the assistance provided while remaining wary of the overly habitual. Consideration of computer assisted composition as a partnering, rather than as a governing, is a positive way of understanding the musical assistance provided by the computer.

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Acoustical, Psychoacoustical and Subjective Assessment of Alvin Lucier's *I am Sitting in a Room*

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This study considers two recordings of *I am Sitting in a Room* in terms of their sound. In physical terms, the spectral content, transfer function between generations, and apparent room reverberation time are examined. Several psychoacoustical algorithms are applied to the sound recordings, in order to rate their sensory qualities. The experimental part of the study involves subjective assessments of a portion of each generation of the two recordings. Three evaluation scales and two emotion scales were used. Relationships between psychoacoustical and subjective measurements are examined. Together, the three parts of the study indicate that the 1980 recording is more appealing and more subtle than the 1970 recording.

Keywords: Alvin Lucier; psychoacoustics; subjective; analysis

Introduction

Composed in 1969-70, *I am Sitting in a Room* is one of Lucier's earliest mature works. Its is highly distinctive musical composition, in that a room is used as the musical instrument. The two most accessible published recordings of this work made by Lucier date from 1970 and 1980 (Lucier 1990, 1997). This paper examines the sound of these recordings in acoustical, psychoacoustical and subjective terms. Thus a work that is most commonly discussed in terms of its production is analysed here in terms of its sound, with an emphasis on perceptual qualities.

The musical score of *I am Sitting in a Room* simply consists of a page or so of text (Lucier 1995). The performer reads aloud the following text, recording their voice in a room of their choice.

> I am Sitting in a Room different to the one you are in now.
>
> I am recording the sound of my speaking voice and I am going to play it back into the room again until the resonant frequencies of the room reinforce themselves so that any semblance of my speech, with perhaps the exception of rhythm, is destroyed.
>
> What you will hear, then, are the natural resonant frequencies of the room articulated by speech.
>
> I regard this activity not so much as a demonstration of a physical fact, but more as a way to smooth out any irregularities my speech might have.

The initial recording is played back into the room using a loudspeaker, and re-recorded with a microphone. As the sound is repeatedly re-recorded thus, the text is progressively overwhelmed by tones produced by the acoustical characteristics of the transducer-room combination. The final sentence of the text refers to Lucier's stutter.

The score is written quite flexibly, with the performer permitted to use multiple rooms, languages and microphone positions. However, the recordings made by Lucier were each made in a single room and in English. The 1970 recording presents 15 generations, while the 1980 recording presents 32 generations.

This paper compares the two recordings, and characterises the evolving sound quality of each. It does not consider the linguistic and cultural meanings of the work. Results are best described as indicative, and generalisations of the findings beyond these two recordings are difficult to make. The paper can be read as a case study of the subjective and psychoacoustical analysis techniques applied.

Acoustical Considerations

If the process described by Lucier's score were executed with a stable and linear system, then we would expect a constant transfer function (and impulse response) between generations. Considering the room acoustics alone, these can be understood as the combined effect of interference between the direct sound and early reflections, and the excitation of room resonances (or modes). If the room is approximately rectangular, axial (one-dimensional) modes should dominate (Cremer and Müller 1978), with each dimension forming a harmonic series of spectral peaks. In typical rooms, the average absorption coefficient varies between surfaces, allowing one set of axial modes to dominate. Interference between the direct sound and early reflections is also characterised by a harmonic series for each pair combination, but multiple pairs contribute to a transfer function in which harmonic series may be obscured (especially when dispersion and absorption as a function of frequency are considered for each path). Whatever the transfer function, it should be reinforced at each generation, so that ultimately a single tone remains, corresponding to the frequency of the transfer function’s highest peak.

The reflections and resonances of the room add to the duration of each generation, because it takes time for the sound to build up and decay. This build-up and decay process also has the effect of smoothing the speech’s temporal envelope.

Aspects of this idealised process are found in Lucier’s recordings. However, the transfer functions between generations are far from constant. The presence of a (non-static) person in the room, movement of transducers, distortion in the electronic stage of the process, the use of variable electronic equalisation, and tape speed anomalies could reasonably contribute to the discrepancies. There is evidence for at least some of these factors in the recordings.

There is little evidence of harmonic series contributing to the 1970 recording transfer functions. By contrast, the 1980 recording has a clear harmonic series based on 37 Hz, with at least 30 harmonics evident starting from the second harmonic. This series is most probably caused by axial modes between opposite walls 4.6 m apart.

Apparent impulse responses were obtained through multiple averaged cross-correlation functions between generations. Apparent reverberation times (following Schroeder (1965)) are of the order of 0.25 s and 0.4 s respectively for the 1970 and 1980 recordings. These reverberation times are reflected by the respective lengthening of the generation durations from Generation 1 to 15 by a factor of 1.02 and 1.05 for the 1970 and 1980 recordings. The contrast in reverberation time is consistent with the hypothesis that room modes have a stronger influence in the latter recording, while the former is more strongly influenced by interference between the direct sound and early reflections.

The later generations of the 1970 recording have a somewhat shrill timbre, caused principally by peaks between 1.5 kHz and 2 kHz. The 1980 recording is dominated by much lower frequencies. The subjective significance of the recordings’ spectra is well-illustrated by Figure 1, where the specific loudness patterns (loudness attributable to specific auditory filters) is plotted against Erb number (a psychoacoustical ‘frequency’ scale derived from the equivalent rectangular bandwidths of auditory filters – see Moore and Glasberg 1987), averaged over the entire duration of each recording. As the area under a specific loudness curve represents the total loudness, the very strong perceptual contributions of frequencies in the vicinity of 1900 Hz and 300 Hz to the 1970 and 1980 recordings respectively is evident.

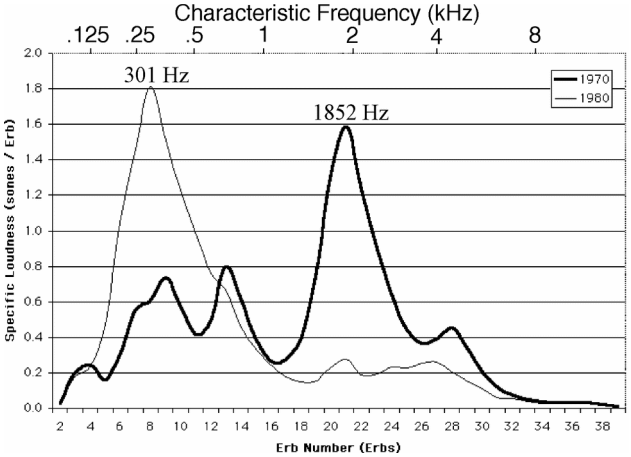


Figure 1. Specific loudness patterns for the two recordings of *I am sitting in a room*, averaged over the entire recording duration. This chart represents the time-averaged stimulation of the auditory system by the two recordings. Auditory filters are distributed linearly on the horizontal axis (based on the Erb number scale), with their characteristic frequencies indicated at the top of the chart.

Psychoacoustical Measures

The two recordings of *I am Sitting in a Room* were analysed using *PsySound* 2.05 (Cabrera 1999a), which applies numerous psychoacoustical models to sound recordings. The recordings were calibrated for an equivalent pressure level of 75 dB(A). Four psychoacoustical measures are illustrated: loudness (following Moore, Glasberg and Baer (1997)), volume (the apparent size of the sound, following Cabrera (1999b)), pitch multiplicity (the number of distinct audible pitches, adapted from Parncutt (1989)) and spectral dissonance (the 'noisiness' of the sound, adapted from Hutchinson & Knopoff (1978)).

The loudness and multiplicity series of the 1970 recording show a loose three-part structure. These measures increase significantly over Generations 1 to 5, stabilising and declining a little in the next five generations. The final five generations see a small recovery in these measures. This structure is accompanied by a continual decline in spectral dissonance, as the bandwidth of the sound narrows, and the noise components of speech are transformed into tones. The measures of sharpness (not illustrated, representing a scale from dull to sharp timbre) and volume yield little of interest.

The 1980 recording measurement series reveal a change in the character of the recording after Generation 16. This appears to be due to the emergence of new sustained resonances above 500 Hz. It seems likely that the conditions of the room or of the electroacoustic system were altered at this point. Regardless of the reason for the change, its consequences are perceptually salient, and add much to the interest in this recording.

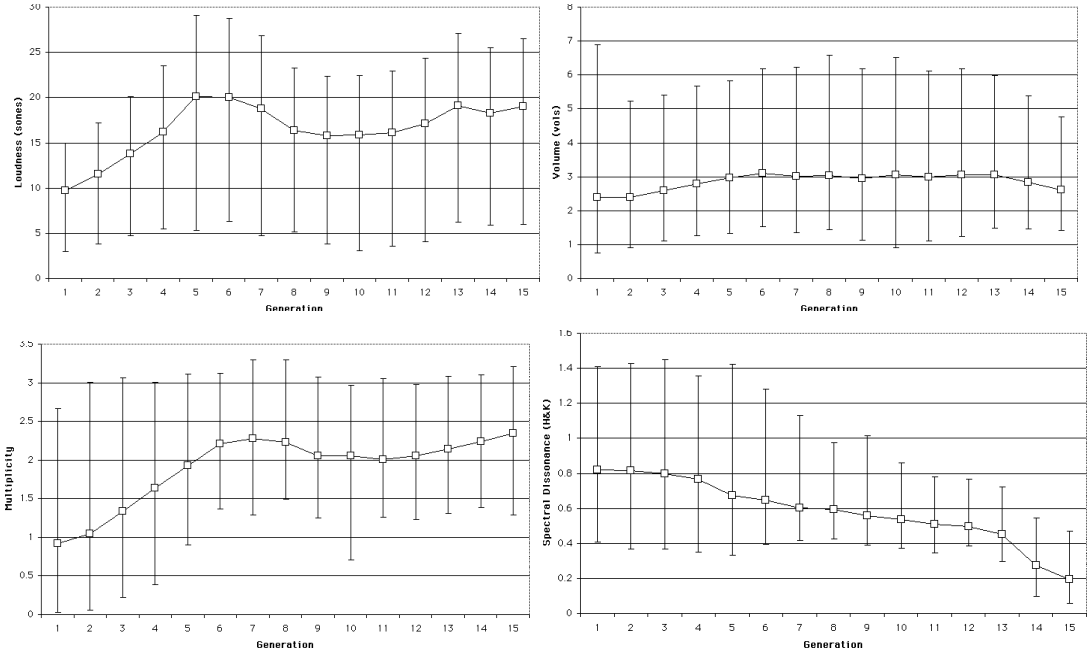


Figure 2. The mean, maximum and minimum loudness (top left), volume (top right) pitch multiplicity (bottom left) and spectral dissonance (bottom right) of the 1970 recording of *I am Sitting in a Room*.

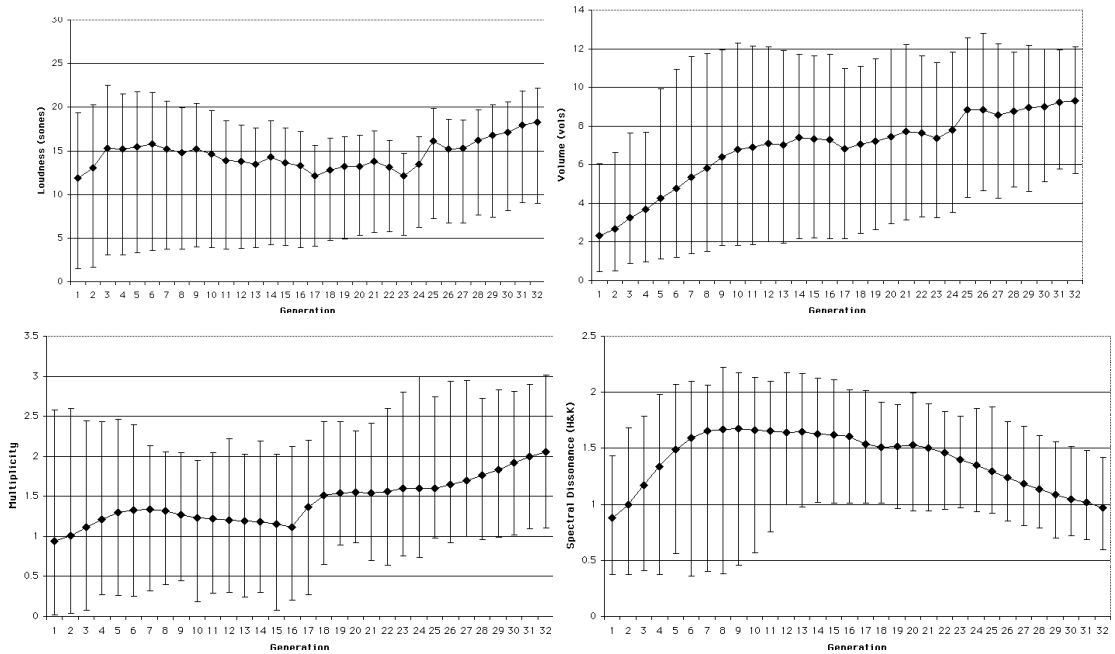


Figure 3. The mean, maximum and minimum loudness (top left), volume (top right) pitch multiplicity (bottom left) and spectral dissonance (bottom right) of the 1980 recording of *I am Sitting in a Room*.

Another loose structural feature can be seen in the way the measures change rapidly initially, and then become more stable. Loudness stabilises at Generation 3, spectral dissonance forms a plateau at Generation 8, and sharpness and volume become more stable at around Generation 10. There is a sense that the process has an initial settling in period, followed by a relatively stable period, which is broken at Generation 16. The few generations following Generation 16 might be characterised as a second settling in period. The final few generations are characterised by increasing loudness, volume, tonalness and multiplicity, and decreasing spectral dissonance.

The variation in mean loudness is much smaller than for the 1970 recording, but the timbral measures of volume and sharpness have large variations with strong trends (sharpness decreases, while volume increases).

When measures are averaged over the entire recording durations, the two recordings have nearly the same loudness (due to the calibration). The 1970 recording is 1.6 times sharper than, and has less than half the volume of, the 1980 version. The 1970 recording has somewhat stronger pitch. The 1980 recording has more spectral dissonance (noisiness), but less tonal dissonance (dissonance between discrete tones).

Subjective Assessments

Thirteen subjects (all with substantial experience in music or audio) rated stimuli consisting of the first sentence of Lucier's text for each generation of the two recordings. The rating scales were bipolar, consisting of (Dis)Like, (Un)Interesting, (Un)Pleasant, Arousal and Valence. The first three are general, intellectual and sensory evaluation scales. The final two are emotion scales, with scale extremes represented by face icons (following Schubert (1999)).

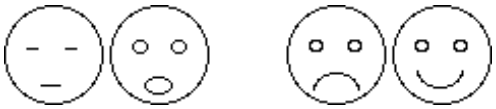


Figure 4. Emotion face icons representing, from left to right, negative arousal, positive arousal, negative valence and positive valence.

The stimuli were presented in the order of the generations, with half the subjects assessing the 1970 recording first. The process of the piece, and the fact that it was a musical composition, were explained to the subjects before the experiment.

The subjective assessments of the sequences of stimuli can be visualised as trajectories in evaluation and emotion spaces (Figures 5 and 6). To reduce evaluation space to two dimensions, we draw on the fact that (Dis)Like ratings are approximately equal to the average of the other two evaluation scales.

Over the first few generations of the 1970 recording, the stimuli become more interesting and a little less pleasant (see Figure 5). Then the bulk of the stimuli are rated as moderately interesting and a little unpleasant. The final three stimuli are the least pleasant and among the most interesting. The trajectory in the emotion space is strong and simple, with Arousal tending to increase over the course of the generations while Valence decreases a little.

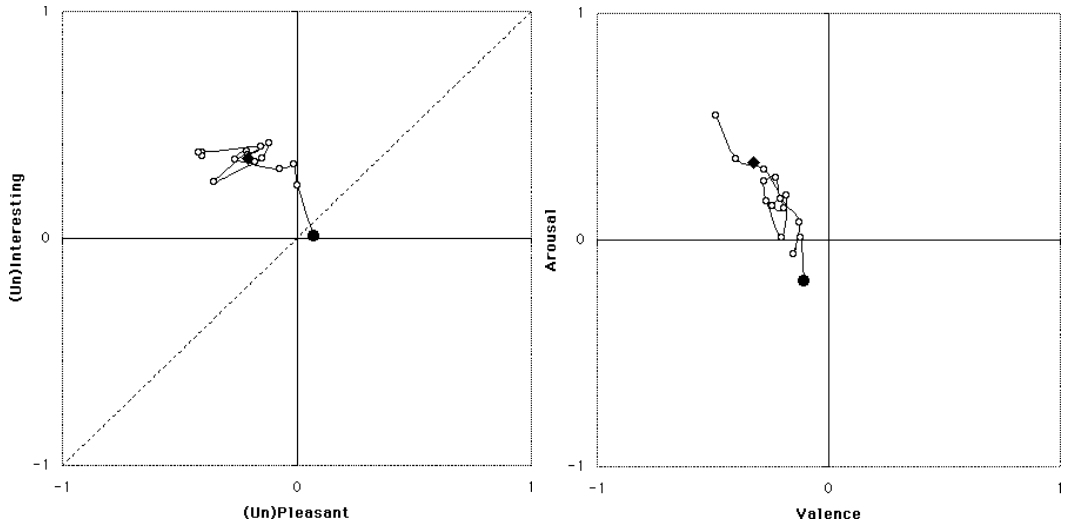


Figure 5. Subjective ratings of the 1970 recording of *I am Sitting in a Room*. The black circle represents Generation 1, and the diamond represents an overall rating made after all generations had been assessed.

Trends are more obscure for the 1980 recording ratings: apart from the beginning, the ratings tend to form a dense cluster instead. The evaluation and emotion trajectories for the 1980 sequence become discernible by plotting a running average, and by enlarging the scale of the charts (Figure 6).

There is a general decrease in pleasantness to Generation 10, accompanied by a smaller increase in interest. Over the remainder of the piece, the generations become increasingly pleasant. Note that the loops in the early twenties are not significant. Interest declines a little to Generation 14, and then increases a little over the remaining generations. There is more variation in pleasantness than interest, and while interest ratings are positive, pleasantness ratings are almost equally positive and negative.

Emotion ratings are very close to neutral, with a bias towards positive Arousal and negative Valence. Most of the variation in the Arousal ratings is not significant. Valence initially declines, but then increases in the final few generations.

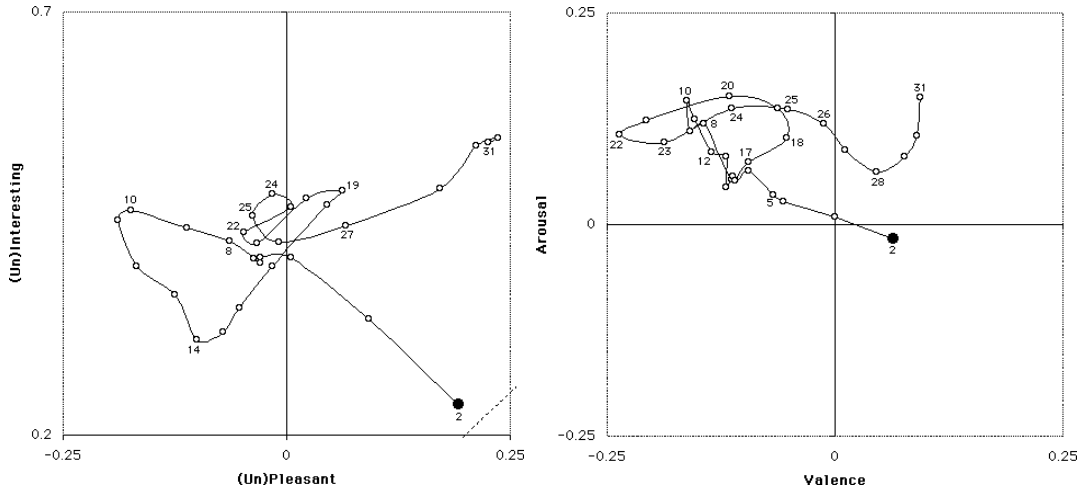


Figure 6. Running average of subjective ratings of the 1980 recording of *I am Sitting in a Room*. Three consecutive generations were used in the running average.

Psychoacoustical measures related to subjective ratings

The problems of serial correlation, low significance and confounded independent variables preclude a detailed analysis of relationships. Nevertheless, the results suffice for broad observations.

As there are simple trends in the 1970 recording ratings (especially for (Un)Pleasant, Arousal and Valence), any psychoacoustical measure that also follows a trend shows a significant correlation with these measures. Loudness, spectral dissonance and multiplicity all have such correlations. The remaining two evaluation scales do not have strong correlations with any psychoacoustical measure (although there is weak evidence of inverted-U relationships with loudness and multiplicity). Correlation coefficients are shown below:

1970 Stimuli	**(Dis)Like**	**(Un)Interesting**	**(Un)Pleasant**	**Arousal**	**Valence**
(Un)Interesting	-0.06				
(Un)Pleasant	**0.74**	-0.52			
Arousal	-0.70	0.61	**-0.91**		
Valence	**0.82**	-0.43	**0.88**	**-0.89**	
Loudness	-0.39	**0.71**	**-0.85**	**0.81**	-0.63
S. Dissonance	**0.75**	-0.57	**0.82**	**-0.84**	**0.91**
Multiplicity	-0.48	**0.74**	**-0.84**	**0.82**	**-0.74**

That Arousal is not related to loudness in the 1980 series almost certainly reflects the much more modest range of loudness therein. However, the ratio of the maximum to minimum loudness for each generation (which can be called loudness contrast), along with multiplicity, is weakly related to the (Un)Interesting ratings. The relationship between spectral dissonance and the (Un)Pleasant and Valence scales is reversed from the 1970 recording.

1980 Stimuli	**(Dis)Like**	**(Un)Interesting**	**(Un)Pleasant**	**Arousal**	**Valence**
(Un)Interesting	0.23				
(Un)Pleasant	**0.83**	0.06			
Arousal	-0.12	0.67	-0.19		
Valence	0.61	-0.02	**0.83**	-0.18	
Loud. Contrast	0.12	**-0.77**	0.15	-0.59	0.24
S. Dissonance	-0.69	0.00	**-0.85**	0.14	**-0.82**
Multiplicity	0.38	**0.76**	0.44	0.51	0.31

Conclusions

We can conclude, on the basis of several findings, that the 1980 recording is much more subtle (from a listener's perspective) than the 1970 recording of *I am sitting in a room*. This is borne out by the more complex patterns found in the psychoacoustical measures and by the much more chaotic but limited subjective assessments. The near-constant Arousal responses reflect the tendency for subjects to favour the less obvious subjective responses when assessing the 1980 sequence. The 1980 sequence involves

markedly lower frequencies (even in Generation 1), is twice as long and involves significant changes in course (whereas the 1970 sequence more or less follows a single course).

Neither recording can be described as emotional. The emotion space trajectory of the 1970 recording is easily understood as Arousal being affected by loudness (which is generally the pattern for studies of this type). In the absence of any positive valence cues, the louder sounds become more annoying, and hence receive more negative Valence ratings. Thus, the 1970 recording is emotionally simple. The 1980 recording emotion ratings are all clustered near the centre of the emotion space, indicating an emotionally neutral (or weak) character. In contrast to the 1970 recording, most variation is in the Valence dimension.

With only two recordings, we cannot generalise about the significance of overall comparisons, including those concerned with the room acoustics. However the more positive evaluations of the 1980 recording, which relies more on room resonance, is consistent with other experiments concerned with the subjective evaluation of sustained room resonance (Cabrera 1997).

Although this analysis of *I am Sitting in a Room* has not addressed any of the symbolic or cultural issues raised by the work, it has shown that the sound itself is rich for analysis, especially when approached from a listener's perspective. The simple progressive structure of the work makes this type of analysis particularly interesting. Clearly similar approaches to analysis could be applied to almost any type of music, as long as the music involved sound. The relevance of such a method depends on the importance of the non-symbolic aspects of sound, and the presence of structural features to give the analysis direction.

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Smart budgeting for a Smart Controller

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The Smart Controller is a portable hardware device that allows performers to create music using Programmable Logic Control. The device can be programmed remotely through the use of a patch editor or Workbench, which is an independent computer application that simulates and communicates with the hardware. The Smart Controller responds to input control voltage and MIDI messages, producing output control voltage and MIDI messages (depending upon the patch). The Smart Controller is a stand alone device -- a powerful, reliable, and compact instrument -- capable of reducing the number of electronic modules required, particularly the requirement of a laptop computer, in a live performance.

Designing and building the Smart Controller may at first appear to require a substantial budget for software and hardware tools. The cost can be reduced significantly when one considers using recycled hardware and low cost software including educational licenses, freeware, and shareware. Subscribing to newsgroups and forums are a viable alternative to expensive customer support, with the added bonus of dispersing the general knowledge base to the wider community. Designing hardware simulators that run on desktop computers – such as Macintosh or PC -- allow the software to be tested before committing to a target hardware.

Keywords: Algorithmic Composition; MIDI Controllers; Recycled Hardware; Software

In 1998, I was involved in two collaborative projects -- the "Laser Harp[1]" and the "Virtual Drum Kit"[2] -- where performers controlled MIDI sound engines using MAX and analogue to MIDI converters. In both cases, analogue signals were converted into MIDI messages, manipulated with an algorithm through MAX, converted back to MIDI, and finally sent to an external sound engine. At least three separate pieces of equipment[3] were required for a performance; each introducing associated software and hardware costs, physical space requirements, and latency in performance. Space restrictions could be reduced by using alternative control voltage (CV) to MIDI converters, such as the I-Cube or some other custom device; however, these devices still require the use of an algorithmic or programmable logic control (PLC) software package on a host computer in order to execute all but the simplest of algorithms or patches. Although one could use a laptop computer, "weight and size isn't everything when it comes to stage (or touring) fitness. The plastic casing and the display on hinges sure don't encourage one to drop the PB [Power Book] from a table."[4] Ultimately, some sort of stand-alone device or "Smart Controller" is required to replace the CV to MIDI converter and the laptop computer, thus allowing a simple interface of controller input to sound engine.

[1] Alex Cockburn. The Laser Harp is an instrument that is performed by cutting laser beams with the body.

[2] Guy Robinson. The Virtual Drum Kit entails a performer playing an invisible drum kit. Each of the drummers arms had two potentiometers that modified a control voltage, thus abstracting the polar coordinates of the drummers arms in space as two MIDI controller values. These coordinates are fed into an adjustable threshold matrix that quantizes each midi message into 1 to 4, and then selects one of sixteen outputs relative to the input coordinates. Each output of the matrix enables a gate that represents a drum.

[3] The Virtual Drum Kit actually had four pieces: SY-77 and SY-99 for analogue to MIDI conversion; Power PC for running MAX; and TG-300 for sound engine.

[4] Sukandar Kartadinata, "hardMax," http://members.xoom.com/Sukandar/hardMAX.html.

I proposed and commenced designing the "Smart Controller," as there were no products currently available that could satisfactorily perform the required functions. Sukandar Kartadinata proposed "hardMax [--] a stand-alone box to run MAX/msp patches;"[5] the project was suspended[6] before I commenced work on the Smart Controller; however, he states that he has resumed work on the project.[7] A release date has not been announced.

In designing the Smart Controller, I am not creating a sound generation unit, but rather an intelligent controller for analogue and MIDI instruments. It responds to input control voltage and MIDI messages, producing output control voltage and MIDI messages (depending upon the patch). The device can be programmed remotely through the use of a patch editor or Workbench, which is an independent computer application that simulates and communicates with the hardware; the workbench, however, is not required during performance. The resultant patches can also be sent to the Smart Controller hardware using MIDI system exclusive messages, which can therefore be saved in the standard MIDI file format. The Smart Controller is a stand alone device -- a powerful, reliable, and compact instrument -- capable of reducing the number of electronic modules required, particularly the requirement of a laptop computer, in a live performance.

Designing and building the Smart Controller may at first appear to require a substantial budget for software and hardware tools. The cost can be reduced significantly when one considers using recycled hardware and low cost software including educational licenses, freeware, and shareware. I decided early in the project to assess where the greatest risks in the project were, and to develop the project focusing upon the software architecture instead of the hardware implementation.[8] Sukandar Kartadinata stated regarding hardMax:

> Development cost: is obviously the biggest stumbling block. Even if a suitable DSP board could be found that would serve as the basic building block and thus minimize hardware development, the software part would still be immense. In any case without source code access to MAX/msp (or at least to some implementation details) it's next to impossible as everything would have to be re-written from scratch and would still suffer from compatibility problems.[9]

It was obvious from this perspective that the viability of the software would have to be resolved before attempting any sort of hardware implementation. "Software is the most expensive thing in the universe….It's cheaper to reuse code. It's smarter to buy, beg, or borrow a module than to code it ourselves."[10] I already had source code for the PLC from Algorithmic Composer.[11] I had intentionally written the code so the PLC engine was not dependent upon the graphical environment, and had low coupling with the operating system primitives. I did, however, have to find a Real Time Operating System (RTOS) to interface the engine with the target hardware. I decided that I would choose the hardware based upon the RTOS features rather than the other way round.

Cameron Shortland introduced me to RTEMS when I was working as a software engineer at Hunter Watertech. He explained to me that RTEMS was both open source and free of cost; the GNU development tools used to build and debug RTEMS were free; the development tools ran on Linux, which

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[5] Ibid.

[6] Sukandar Kartadinata , Cycling '74 Discussions ->MAX-MSP -> Controllers -> Smart Controller, 7 Sep 2000, http://www.synthesisters.com/cgi-bin/WebX?13@174.1y06asTPaIY^11@.ee6bdbe/9

[7] Sukandar Kartadinata, email to Angelo Fraietta, 5 June 2001.

[8] Craig Larman, *Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design*, (New Jersey: Prentice Hall, 1998), 434.

[9] Sukandar Kartadinata, "hardMax."

[10] Jack G. Ganssle, "Open Source vs. Proprietry," *Embedded Systems Programming* 13 (Dec 2000), 189. Ironically, his argument was against the use of open source software.

[11] Angelo Fraietta, "Algorithmic Composer", Honours thesis, University of Western Sydney, 1998.

was also free. Additionally, at that period of our employment, we were working on telemetry firmware based upon the i386ex CPU. We were using the ATI Nucleus RTOS and simulating the telemetry hardware an old 386 PC before we had any real hardware to work with. RTEMS supported both the i386ex and the PC386. I decided to consider RTEMS as my operating system and the i386ex as my target hardware device because I would be able to work on the software and simulate the hardware with a PC.

My first step was to set up a suitable host network at home where I could run Windows and Linux concurrently, which meant that I required at least two computers. I received a broken Pentium 75 computer from a friend, John Turon, as payment for fixing his other computer[12] and used the parts to build another computer. I obtained a Kingston Turbochip 400 for $237 over the Internet and a second-hand motherboard from the computer fair for $50. This enabled me to use the 64MB of RAM that I already had in a 400 MHz computer. I installed Red Hat Linux 6.2 (free) on my original computer, a Pentium 133, connected it to my Windows Machine using Ethernet, and used SAMBA[13] (also free) to copy files between the two computers. I was also able to control the Linux machine using telnet from the Windows machine, thus removing the requirement for a monitor on the Linux machine. This setup enabled me to build the RTEMS source on the Linux machine while concurrently working at the Windows machine. I later received a broken DX 386 computer from a friend, David Gibbins, who said that I could use it if I could fix it – he also insisted that he pay me to fix it. The hard drive was intermittent and so I told him it was not worth his money to fix it -- so he let me keep it (I provided him with a CDROM of the data on the drive). I replaced the hard drive with the one from the Pentium 75, installed Linux, and use a free ftp client, WS_FTP, to transfer the Smart Controller binaries that are built for RTEMS to the PC 386. The computer now uses a multi-boot menu to boot into Linux or any of the RTEMS binaries.

I found out through the RTEMS newsgroup that I should boot the computer into RTEMS using GRUB.[14] After creating the GRUB boot floppies, I successfully booted the Hello World program on my PC. I removed or re-defined all the Windows specific calls in the Algorithmic Composer source (now called the Smart Controller engine source), and built it for RTEMS using the GNU development tools. I booted the computer with GRUB and successfully created objects via the computer keyboard using a simple menu interface.

I was successfully able to build the RTEMS source with the Smart Controller engine source on the Linux machine and run it on a PC within two months of starting the project. This was primarily due to the fantastic support of the RTEMS user group, which was truly shown when I was testing the soft-floating point capabilities of the RTEMS tool-sets. The use of the DX 386 was significant as a simulator as it did not have a math co-processor, which enabled me to test the floating point libraries that would be used in the final target. I was initially unable to run the binaries on the 386 (I had been testing them on a Pentium machine) and posted a message through the mailing list.[15] I was able to determine (after being notified through the mailing list that the problem was a co-processor error) that I could stop the fault by simple placing a co-processor into the 386 computer (fortunately, I had had a co-processor in an old IBM 386 that was no longer serviceable). What eventuated over a period of thirty-six days was a combined effort from the RTEMS users to find and fix the problem, which resulted in thirty-eight email messages. Ralf Corsepius found the problem, but the staggering fact is that most of the people working on the RTEMS project are volunteers.[16] OAR Corporation, the distributors of RTEMS, also provides paid support -- if this was the level of the free support, I find it hard to imagine how good the paid support must be.

[12] All I did to fix his computer was to run Norton's Disk Doctor.

[13] SAMBA comes with Linux Red Hat 6.2.

[14] Mahesh revuri@seeyes.co.in , Re: hello_world_c,
http://www.oarcorp.com/rtems/maillistArchives/rtems-users/2000/august/msg00022.html

[15] Angelo Fraietta, Soft Float for PC386, 23 Jan 2001. (Mailing list archive is missing)

[16] Joel Sherrill, Re: Soft Float for PC386, 28 Feb 2001. (Mailing list archive is missing)

RTEMS was originally developed for the US Army, but was released for non-military uses later.[17] When asked why RTEMS was given away instead of being sold, Mark Johannes of OAR stated "It was a natural fit to release RTEMS as open source to leverage the environment that promotes the further development and utilization of the product."[18]

During the development, I have used many other software packages that were obtained either as freeware, shareware or with an academic license. A listing of the software has been provided at the end of this paper. One shareware package, "Hex Workshop Hex Editor,"[19] was so good that I actually decided to pay for it. Hex Workshop enabled me to find and edit the binary file that was crashing the 386. I found the hex values of the floating point instructions by disassembling the binary using "objdump" (free with Linux) to find the binary file position that had the floating point instructions, and then replaced the floating point instructions with NOP instructions[20] using Hex Workshop.

Some software tools have a shareware version with limited functionality, enabling the user to try (or get hooked on) the product. When the user finds that he or she requires the full version, they are forced to pay for the full version. I decided that I would use a computer-aided software engineering (CASE) tool to generate code from class diagrams, which in turn would facilitate concurrent code and documentation synchronization. I requested and received a copy of Rhapsody 2.3 Modeler from I-Logix through their Web page http://www.ilogix.com. I read about this free modeler from an advertisement in Embedded Systems Programming Magazine[21]. The product is in direct competition with Rational Rose, and so I-Logix provides it for free in order to take some of the market from Rational. I already had a copy of Rational Rose Student Edition[22], so the issue was not so much the price, but which product better suited my needs. Although a complete comparison between the products is beyond the scope of this paper, I chose Rational Rose mainly because it enabled me to edit code both from within and outside of the CASE tool. Additionally, *Visual Modeling with Rational Rose and UML*,[23] gave an in-depth description of how to use the CASE tool effectively, with a section on code generation and reverse engineering.

Designing hardware simulators that run on desktop computers – such as Macintosh or PC -- allow the software to be tested before committing to a target hardware device. I have been able to run and debug all of the non platform specific code on my PC at home using the compilers that I am already familiar with (Borland BC 5.02). After building and running the code on the PC, I build the code for RTEMS and run the binary on the 386. This has had a twofold effect: the code is compiled using completely separate compilers, thus facilitating platform independent code; and I can always run the code and know where it is not faulty in the embedded target. One area where this approach has been particularly successful was during the development of the workbench and editor communication protocol. I managed to develop and test the interface in two stages: implement a GUI and perform all engine queries and manipulations through a single function call; and perform the manipulations through the communications ports on the computers. In the first instance, I created a console driven menu to create and delete objects through the upper layers of the interface protocol. After succeeding, I modified the Algorithmic Composer GUI code and hooked it into the interface. The success of this stage proved that I could interface between the GUI and the engine by passing parameters between the two sections using a character buffer. This meant that interfacing the two though a serial cable was possible by implementing a lower layer, which could be

[17] Joel Sherrill, Re: History of RTEMS, http://www.oarcorp.com/rtems/maillistArchives/rtems-users/2001/march/msg00033.html

[18] Mark Johannes, Re: History of RTEMS, http://www.oarcorp.com/rtems/maillistArchives/rtems-users/2001/march/msg00032.html

[19] http://www.bpsoft.com/ordering/.

[20] Barry Brey, *Embedded Controllers: 80186, 80188, and 80386EX* (New Jersey: Prentice Hall, 1998), 575.

[21] *Embedded Systems Programming* 13 no 8 (2000), 41. The advertisement states "There is nothing Rational about paying for UML Modeling."

[22] http://www.rational.com/products/rose/tryit/stud_edition.jtmpl

[23] Terry Quatrani, *Visual Modeling with Rational Rose and UML* (Reading: Addison-Wesley, 1998).

done separate from the application. I designed the lower layer and tested it with a driver application through the Windows communications ports. I then implemented the lower layers into the Smart Controller code. I was able to then control the engine running on one computer with a GUI running on a different computer through a serial cable. This has proved that the workbench on a desktop computer can control the engine remotely. This step was vital before attempting to embed the engine software because I know that an inability to communicate with the embedded device is not due to the interface protocol.
Developing the Smart Controller is more than just developing a target device – it is developing a methodology whereby I do not have to always put my hand into my pocket to develop software systems. When I do, however, I know I am getting the best value for my (Australian) dollar.

Software Packages Used

CVS -- Concurrent Versions System. Management system for controlling and tracking source code modifications. http://www.gnu.org/software/cvs/. Cost: free.

Cygwin. Provides a Unix/Linux environment when developing in Windows. http://www.redhat.com/products/support/cygwin/. Cost: free.

GNU Emacs. A powerful text editor available for most platforms. http://www.gnu.org/software/emacs/. Cost: free.

GNU GRUB. Multi-boot boot loader. GNU GRUB is derived from GRUB (GRand Unified Bootloader), which was originally designed and implemented by Erich Stefan Boleyn. http://www.gnu.org/software/grub/. Cost:free

Hex Workshop Hex Editor. Hex editor for modifying binary files. http://www.bpsoft.com/ordering/. Cost: $US24.95 academic.

Rational Rose Student Edition – CASE tool limited to thirty classes and three packages. http://www.rational.com/products/rose/tryit/stud_edition.jtmpl. Cost Free.

Red Hat Linux 6.2. Linux OS obtained from Ray Kiefel at Hunter Watertech. http://www.redhat.com/products/software/linux/. Cost: free under the GNU GENERAL PUBLIC LICENSE.

RTEMS – Real Time Executive for Multiprocessor Systems. GNU tool-sets are also available. http://www.gnatrtems.com/RTEMS/rtems.html. Cost: free

WS_FTP Limited Edition – File transfer client. http://www.ipswitch.com/. Cost: free for academic use.

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RTEMS Mailing Archive. http://www.oarcorp.com/rtems/maillistArchives

A survey of recent electro-acoustic music from Latin America

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Abstract

A concert-talk featuring some recent cutting-edge electroacoustic and computer music pieces by Latinamerican composers covering different trends and approaches, be it aesthetical, technological, compositional, etc. There will be references to historical landmarks in Latinamerican ea/cm and how they influenced today's production.

These are some of the pieces I plan to perform:

"Due giorni dopo" by Ricardo Dal Farra (Argentina) 3:00

"SC" by Martin Fumarola (Argentina) 3:00

"La beauté indiscrète d'une note violette" by Jorge Antunes (Brazil) 2:00

"Crystal Ball" by Jacky Schreiber (Venezuela) 5:00

"Arena" by Francesca Ancarola (Chile) 5:00

Martín Fumarola is an Argentinian composer with a special interest in electroacoustic and computer music. His electroacoustic music is regularly performed in international festivals and concert series in Europe, the USA, Canada, Latin America, and Australasia. He is currently an Associate Composer of the LEIM of the National University of Córdoba (Argentina).

http://www.geocities.com/maralefu
http://www.electrocd.com/bio.e/fumarola_ma.html

Studio Time = Money

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NSW 2011

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Abstract

In this talk, I argue that an analysis of how monetary relations define and transform musical practices is essential to an understanding of current directions in contemporary music. I will examine the enduring problem of monetary pressures on musicians and the significance of money for musical production, and consider how the force of monetary relations has shaped on-going debates among musicians about the value of their art. I will also consider how the advent of analogue and digital processes-in both the realms of popular and experimental music---has reinforced the cycles of optimism and pessimism that defines much cultural criticism. At issue will be the role of the recording studio in contemporary music, particularly as a consequence of mass commercial recording, and the social myths and rites that inform and orient current studio practices. Of specific concern will be how musicians use their studio time, who pays for it when they do, and at what cost to themselves and to music. Thus, the overall focus of this presentation is on the social value of contemporary music acts.

Singing Me Softly...

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Abstract

This short paper will consider the Oracular powers of Sound in the prediction and construction of imaginative Utopic (and Dystopic) futures - in effect a pre-technological form of Virtual Reality. I will discuss the traditional fusion of site, sculpture, speech and belief systems, which characterise Oracles and then focus upon the relationship of Sound to Political power and territorial conflict. The final section of the paper will examine the re-emergence of Sound as a spatialising metaphor and agent in Cinema and in immersive Virtual Reality systems.

Dr. Nigel Helyer is an Australian Sculptor and Sound-Artist with an International reputation for his architectural scale sound installations, public art projects and experimental radio broadcasts.

His initial formation as a sculptor has provided him with an orientation insistent upon an experiential, rather than a representational means of expression - one in which the body, its sensorium and its memory, is invoked as the primary location of meaning. This has drawn him toward an area of research and practice which seeks to critique and develop a nexus of relationships between the body, sound and public space.

He is currently the Artist in Residence at Lake Technology (Sydney) where he is undertaking Research and Development of the virtual audio reality project “Sonic Landscapes”. He is an International commissioner for the “SoundCulture” organisation, a consultant to the “ReSonance” sound-art festival (Seoul 2001) and a member of the New Media Board of the Australia Council.

http://www.sonicobjects.com
http://www.laketechnology.com

Yasunao Tone's stuttering CDs and extreme digital audio

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Abstract

Yasunao Tone is a composer who has been a much neglected part of the Fluxus movement. At the centre of the moment in Tokyo he was part of a group called Group Ongaku in the 1960s as well as Team Random, the first computer art group in Japan. In June 1972 he moved to American where he currently resides. Since moving to America he has worked with the Merce Cunningham Dance Company and had solo concerts in venues such as The Kitchen, Experimental Intermedia, Roulette, P.S. 1, and The Knitting Factory. In 2000 he received an honorary mention in the Prix Ars Electronica for the work *Wounded Man'yo 2/2000*.

Tone has come to be known for his work with *wounded* CDs which he has used in composition since 1985 when he added scotch taped to a Debussy disc and set it off skipping and jumping through the piece. For *Wounded Man'yo 2/2000* he used a previously computer generated composition burnt to CD-R and then *wounded* to produce a live remix via the CD-player. This method of composition has its history in Tone's early use of indeterminacy and his event works for Fluxus related projects. The paper will explore this history and contextualise his contemporary practice in terms of 'digital audio.'

Spatialised sound: the listener's perspective[1]

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This paper compares the listener's perceived differences, in terms of space, between sound created by acoustic instruments and loudspeakers. The argument is made that, unlike acoustic instruments, sound from loudspeakers is not perceived as existing in the same personal space as that of the listener. Given that the large majority of music is now heard via loudspeakers, the paper discusses the possible implications in cultural terms. Following on from this is a discussion of some of the potential solutions to the spatial limitations of loudspeakers that have been developed. These include the creation of surround sound systems and virtual sound illusions via signal processing. Conclusions are made as to possible developments that could occur in the area of spatialised sound via loudspeaker playback.

Keywords: Sound Spatialisation, Loudspeakers, Ambisonic, Acousmatic Music, Sound Localisation

Introduction

Electronic music has, in the large majority of cases, only one actual instrument – the loudspeaker. This instrument is remarkable in its ability to create an enormous range of sounds. In fact so much so, that loudspeakers are often wrongly assumed to be able to create, or reproduce, any sound at all. For musicians and composers, loudspeakers can become conceptually invisible in the creative chain. This is despite the fact that loudspeakers do have limitations, particularly in relation to spatial imaging and, that these limitations strongly influence the type of music created for them.

- If we think about the spatial element in sound from the listener's perspective, then there are four main factors to consider:
- the sound itself,
- the environment in which the sound is transmitted,
- the spatial cues encoded in the sound[2],
- the correlation between spatial cues and the listener's ability to identify and position the sound source.

These factors affect the way a listener will perceive the spatial element of the sound and how they might relate this to a musical context. In considering this, it is first necessary to consider the sound sources that are under discussion here.

All physical objects, such as musical instruments and loudspeakers, must operate in an air-filled space (referred to here as *acoustic space*). As electronic music is almost completely reliant on the loudspeaker, it is useful to understand the relationship of the four elements of spatiality mentioned above, to the behaviour of loudspeakers in an acoustic space.

Furthermore, an understanding of the differences between musical instruments and loudspeakers, in these terms, seems desirable as many of the conventions and culturally-based interpretations of music stem from our tradition of music played by instruments.

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[1] This paper is an adapted version of sections from the masters thesis "An Approach to Spatio-temporal Patterning in Computer Music" Peter McIlwain, 2000.

[2] For a detailed discussion of sound localisation, see Kendall (1995).

One of the major differences between musical instruments and loudspeakers is that loudspeakers are able to create spatial cues from a different environment to that of the acoustic space of the listener. In the case of headphones, these spatial cues are, in the main, not modified by the listener's acoustic space. Here the listener can become immersed in a different acoustic space to the one in which they are situated. This is not a real space, in that it is not the result of the physical environment in which the listener is in. For this reason these spaces are described here as *virtual spaces*. In addition to the discussion outlined above, this paper will discuss the implications for music presented in virtual space compared to music that occurs in the physical environment.

A Comparison Between Loudspeakers and Musical Instruments

As discussed above, both loudspeakers and musical instruments operate in acoustic space. Their interaction with acoustic space, and the way in which the listener decodes the spatial information from these two types of sound sources, can be different. The spatial characteristics of sound coming from an acoustic instrument tends to be perceived by the listener in such a way that it is directly linked with the acoustic space in which they and the instrument reside. In other words, the sound is perceived as being in the listener's personal space. On the other hand, sound from loudspeakers tends to be heard as coming from somewhere else, or at least as not being part of the listener's acoustic environment. This perceptual difference may be caused by three factors:

- the difference between the ways in which the two types of instruments radiate sound,
- the fact that the playback of recorded sound through loudspeakers carries spatial information from another acoustic space[3], whereas acoustic instruments do not, and,
- the difficulty that listeners may have in creating imagined sound sources, and therefore sound loci, for electronically-generated sounds.

Acoustic instruments generally project sound in all directions. Their radiation patterns are typically complex as the pattern changes depending on the frequency that is emitted by the instrument. It is this complexity, together with the resultant reverberation caused by the interaction of the environment and the projected sound, that gives listeners clues as to the identity of the sound and its position in space (Roads, 1996).

Loudspeakers, on the other hand, radiate sound in a forward projecting *radiation pattern*. In the case of the loudspeaker, the listener is mainly hearing (in optimal listening conditions) the direct sound coming from the speaker. In contrast, when listening to an acoustic instrument, the sound of the instrument becomes part of the acoustic space in which it is created. This is due to the fact that the listener is hearing direct sound, as well as the complex interaction of that sound with the acoustic environment. On the other hand, complex interactions between the environment and sound coming from loudspeakers are more likely to distort, or interfere with, the intended sound. Furthermore, radiated sound from loudspeakers carries fewer clues as to the identity and localisation of the sound. For this reason locatisation cues are often included in the audio signal (such as artificial reverberation). However this solution gives rise to the second problem where the sound of a virtual space is superimposed upon the acoustic space.

A third factor comes into play when projecting sound over loudspeakers which is related to ability of a listener to create what Wishart (1996, pp 129 –161) calls *a sound landscape*. In his extensive discussion on the spatial properties of sound he states that the mental apparatus concerned with hearing is predisposed to allocating sounds to sound sources. This happens naturally in the performance of music played by acoustic instruments where the sounds are sourced to their physical origin. Listeners are able to do this by correlating the direct and indirect sound of the instrument with the visual stimulus of the performers playing.

[3] An exception to this is recordings that are made with microphones in the same space and position as the loudspeakers that playback the recording.

If however a recorded performance of acoustic instruments is played over loudspeakers the physical location of the sound is the loudspeakers themselves. In this instance, according to Wishart, the listener imagines the spatial and acoustic environment, or sound landscape, of the recording. The listener is able to do this because the sounds, and the mechanisms that generated them, are known. This is important because the listener is able to relate the behaviour of the sound of the instrument with that of the environment (such as reverberation or contextualising sound cues that may help to determine the nature of the acoustic space).
Sounds from synthesisers, on the other hand, are harder (or even impossible) to source, even in an imaginary sense. This is because they do not have a physical source. A listener therefore, may not be able to imagine a landscape for synthesised sounds as they have no physical origin. This can also apply to recorded sounds that are difficult to source or ambiguous as to their identity. For this reason, sounds of this type projected over loudspeakers may sound detached from the listener's acoustic space and even their imagined spaces.

Virtual and Acoustic Spaces with Stereo Loudspeakers

The difference in the spatial qualities of the two types of instruments is particularly marked where stereo sound recordings are played over stereo loudspeaker systems. These recordings carry the spatial characteristics of the environment in which they where recorded. As discussed above this sound exists in a virtual space. When the recording is played, it must be transmitted through the acoustic environment of the listener. This can result in a distortion, or even collapse of, the virtual space, as the two different spaces are overlayed. In order to avoid this, loudspeakers are often utilised in such a way as to negate the acoustic properties of the space in which they are in. One way that this is done is use loudspeakers in acoustic spaces that have high acoustic absorption characteristics and thus minimal impact as an acoustic space.

Listening to a recording produced by loudspeakers is the perceptual equivalent of looking at photographs. The sound is a kind of imported reality or the illusion of reality (MacDonald, 1995). If a sound is imported into a listener's 'reality' and is therefore not part of their perceived acoustic space, then, by inference, it must come from somewhere else.

This observation, in relation to loudspeakers, has been noted by Moore (1989) in his paper "Spatialization of Sounds over Loudspeakers". Here he describes the experience of listening to loudspeakers as perceptually equivalent to a situation in which a listener, in a room, is hearing a sound through a hole in the wall. An everyday example of this is when a listener in a room opens a window. The sound that is heard coming through the window is from another acoustic environment and is heard as such.

It is important to note here that Moore's observations are not just supposition. He is able to show that his model of perceived spatiality in relation to loudspeakers works by giving an algorithm that can create spatial effects. The algorithm works by calculating differences in onset times and relative amplitudes for paths travelled by sound through two "holes in a wall" (see Figure 1).

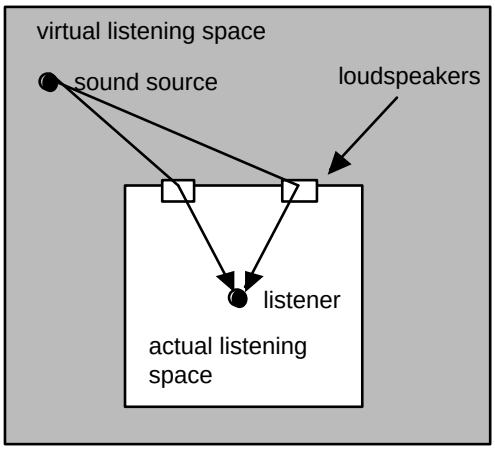


Figure 1. Moore's algorithm for spatialiation that makes a distinction between the ‘virtual’ and ‘actual’ listening spaces.

The Spatial Illusion and Listening Culture

The development of audio recording and reproduction technology introduced a radical change in the perception of music. It was now possible to listen to music that was not created in the same time and space as that of the listener. The modern experience tends to take this fact for granted even though (or possibly because) it is now the case that most music is heard through loudspeakers. The tendency to relate, at an unconscious level, loudspeakers as the producers (or reproducers) of “sound which comes from somewhere else”, has become part of the culture of listening to music produced by electronic means.

This culture now embraces not only spatial illusions which are intended to reproduce real acoustic spaces, but illusions of space which would not be possible in the real world. An example of this is exemplified in one approach to the recording of orchestras. Here a balance between the various instruments is created which is not necessarily intended to represent what is heard in the concert hall. Instead the balance is created such that all of the instruments can be heard clearly. Instrumental parts that might be masked when heard in the concert hall, due to the limitations of the power of the instruments concerned, can be compensated for in the recording process. This type of balance could only be achieved in a real world equivalent if the listener were able to be in several places at the same time.

Popular music, which is often created in the recording studio, commonly uses unreal acoustic spatial illusions. For example, a vocal line can be treated in such a way that it appears that the singer is enormous in size (as a spatial illusion) compared to the backing ensemble. This is done by the use of a combination of reverberation and artificial balance. Just like the larger than life presence that movie stars have on the cinema screen, popular singers in recordings can have a god-like presence. They can sing with a power not possible in the real world and, due to recording and broadcasting technology, defy time and space.

These illusions cause music to be heard by the listener as distinct from the mundane or everyday listening experience. Indeed the spatial imagery commonly found in popular recordings creates a spatial context for music that is unreal or even surreal. Listeners accept the unreality of music heard through the stereo sound system in the same way in which they are able to accept the unreality of a puppet show – by a kind of aural suspension of disbelief. Kendall (1995) describes recorded music as a

> “learned cultural form that we usually take to be a mediated approximation to direct experience”.

Unreal illusions of space can provide many creative approaches to the composition of music. They may, however, reinforce an unconscious association in which the sound that is being heard is distinct from the mundane and removed from the listener’s personal space. This facilitates the commodification of music

and sound by the listener. The recording industry can be seen as producing objects that are consumed the same way as any other mass-produced good — machine perfect and not able to be made by individuals. This can give rise to the speculation that the everyday immersive experience of sound has diminished as an aesthetic experience for a large number of people. Support for this can be seen it the fact that acoustic spaces, such as shops, cars, workplaces and homes are increasingly filled with the sound of radios and other playback equipment. Here everyday experience is masked by a spatial illusion.

Multi-speaker Systems and Acoustic Space

Since the 1950's composers have experimented with multi-speakers systems. These experiments have given rise to approaches that utilise the acoustic space in which the loudspeakers are in. An early example of this is Karlheinz Stockhausen's *Gesang der Jünglinge* (1956) which is a piece for tape consisting of 5 tracks distributed to 5 loudspeakers according to a predetermined plan (Stockhausen, 1961). Another notable piece from the same period is *Williams Mix* (1953) by John Cage, a work for eight channels of audio. Here the spatial distribution is determined by Cage's chance operations (Chadabe, 1997 pp. 56 -58).

In 1974, François Bayle created the *Acousmonium*, which consisted of an orchestra of 80 loudspeakers. Bayle placed the loudspeakers on a stage arranged in a manner similar to that of an orchestra in that the instruments where placed according to their range, power and other projecting qualities (Bayle, 1986). This method of presentation has become a popular model for composers utilising multi channel sound.

Another approach is seen in the *Moving Sound Creatures* created by Felix Hess in 1987. Hess created 24 simple robots that were mobile and produced sounds. Together the robots create a kind of dance that results in a continually changing spatial distribution of sound (Chadabe, 1997).

The works discussed above overcome the problem of directionality in loudspeakers by using multiple speakers, thereby projecting the sound in a way that is similar to the way in which ensembles of instruments do. In addition they also overcome the problem of superimposed spatial imagining between virtual and acoustic spaces as there is no intention to create a virtual spatial illusion in the way that occurs with the stereo sound system. Instead space is created in the acoustic space of the performance by the use of multiple speaker arrays and the animation of sounds occurring in those arrays.

The development of multi-speaker techniques has given rise to an approach called *Sound Diffusion*. Here composers such as Harrison[4] distribute a limited number of channels, via a mixing console, to a larger number of loudspeakers. This method of spatial distribution has been expanded further with the introduction of software that automates the process of distributing the sound. An example of this is the ABControl software interface that controls the Richmond AudioBox which was recently used at the Sonic Residues concerts in Melbourne last year[5].

While this approach affords many options for composers working in this area it can fall short of creating a convincing sonic landscape. If Moore's concept of spatial imaging in loudspeakers is taken into account this criticism becomes apparent. In Moore's model, loudspeakers function like windows that open up to a world of sound outside the space in which the speakers occupy. Therefore a single sound that is assigned to one loudspeaker in a multi-speaker array is not convincing unless it is distributed to all of the loudspeakers with corresponding alterations in onset times and intensity that allow for the difference in paths taken for sound to reach the respective loudspeaker windows.

The solution to these problems requires the creation of virtual acoustic spaces. There is now a large body of literature on this subject that outlines a range of approaches to the creation of virtual space. However

[4] During the mid 1980's Jonty Harrison and colleagues at the Birmingham University developed BEAST (Birmingham Electro Acoustic Sound Theatre) which utilises a complex assortment of loudspeakers for sound diffusion (Chadabe 1997 p. 132).

[5] For further information about the ABControl and the Richmond AudioBox software see: http://www.thirdmonk.com

this literature tends to be highly technical making it somewhat unapproachable for many composers. For this reason the discussion here will cover some of the approaches in conceptual terms oriented towards a compositional perspective.

Virtual Spaces and Loudspeakers

Virtual spaces are created by playing sounds which have spatial cues present in the signals. In the case of the stereo sound system, this is done by recording sound with two microphones that record spatial cues created by the environment as part of the signal. This method can be extended to multi-channel recording when more than two microphones are used. An example of this is the Ambisonic system for the recording and playback of sound. This system not only records right and left information (using the coincident pair method used in stereo) but also front and back, and height[6].

Another method is to subject a sound to signal processing that modifies the signal to include spatial cues. An early example of this, in which a computer was used to simulate moving sounds in a quadraphonic sound system, was developed from 1966 onwards by John Chowning. Here Chowning developed a computer program that calculates a range of spatial cues (such as relative energy levels, reverberation and doppler shifts) necessary for moving sounds within a virtual space (Chowning, 1971). After Chowning's work, a wide range of approaches, using the signal processing capabilities of computers to create virtual spaces, have been developed.

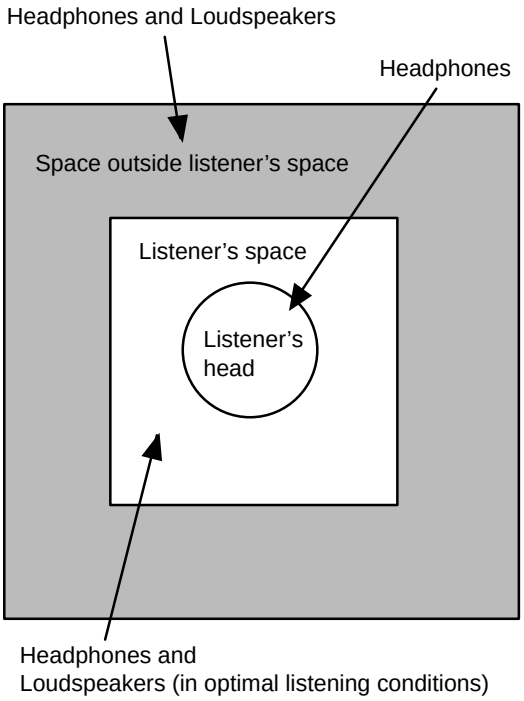


Figure 2. Three possible spaces that can be created using either headphones or loudspeakers.

[6] See Malham (1998) for an introduction to Ambisonic technology.

Virtual spaces can be categorised into three basic types (see Figure 2). They are the space:

- outside the listener's acoustic space,
- inside the listener's acoustic space, and,
- inside the listener's head.

The boundaries of these spaces are determined by the position of the loudspeakers. In the first and second categories, the space exists either outside or inside the boundary defined by the positions of the loudspeakers. In the third case, it is delimited by the position of the loudspeakers in the headphones.

The first category is addressed by Moore's system for spatialising sound. Moore's algorithm relies on the idea that loudspeakers are windows into a space outside the listener's space. It is a robust method because it not distorted by the acoustic space. This is because the spatial illusion is heard outside, and therefore distinct from, the acoustic space. Moore's approach however, is not able to create illusions for sounds positioned inside the listener's acoustic space.

Chowning's approach, and similar approaches after Chowning, attempt to position and move sounds within the listener's space[7] (the second category described here). When signals processed in this way are played over loudspeakers however the problem of the overlay of virtual and acoustic spaces becomes a factor. For this reason, loudspeaker systems, such as the speaker dome developed at ACAT, must be used in an environment in which reflected sound is kept to a minimum (Vennonen, 1995).

This problem is avoided if loudspeakers are replaced with headphones. However this gives rise to another problem. If the extent of the virtual space is now between the loudspeakers, then this space must be, to the listener, inside the head. While there is some novelty in hearing sounds in this way (although at times the sensation can be somewhat disconcerting) there is still a failure in representing sound as occurring in the acoustic space of the listener. If the sounds are positioned so that they are perceived outside the bounds of the headphones then we are returned to Moore's model of spatialisation.

In terms of an immersive sound field, headphones are problematic for several reasons (see Kendall, 1995 pp. 37-39). One of these reasons is that if the listener moves his or her head the virtual image will move also. The result of this correlation of movement is the conclusion that the sound must be inside that head.

With the advent of the DVD standard for the domestic market (Dennis, 2000) there has been a rapid growth in software that performs spatial processing[8]. A survey of this new growth in electronic music is well beyond the scope of this paper.

Space as a Musical Parameter

As discussed above, there have been a number of approaches to incorporating spatiality into the distribution process. However, in the compositional process, spatiallity has tended to be a secondary consideration, or at least a consideration that is imposed or overlayed after the other compositional parameters have been set. It therefore seems timely that the development of techniques for the composition of spatial parameters be addressed.

An important contribution towards incorporating spatiality into the compositional process can be found in the writings of Wishart on the subject of spatial motion (1996, pp 191 – 235). Here Wishart catalogues a wide range of trajectories that could be used for moving sounds in a two dimensional plane. While this kind of analysis is useful, Wishart does not mention how his catalogue of trajectories might have a

[7] These approaches can also position sounds outside this space.

[8] Examples of this can be seen in the plugins being used by audio and MIDI sequencing software such as Digital Performer (Mark of the Unicorn, 2001). These programs have incorporated mixing facilities that enable mastering to DVD audio specifications.

correlation with other musical parameters. Furthermore, he stops short of discussing compositional methods or approaches to the long-term organisation of the trajectories he presents.

An approach that extends the idea of space as a compositional parameter is found in the author's SNet software (McIlwain & Pietsch, 1996). This approach attempts to incorporate the spatial parameter in such a way that it becomes intrinsic to the compositional process. Here the spatial parameter is considered in the same manner as are the commonly considered parameters of frequency, amplitude and time. The consideration of these three parameters in the compositional process is usually interrelational, that is, depending on context, a composer will base decisions about one of these parameters in the context of the other two. This is done in SNet by the use of a nodal network in which all of the parameters influence, in an interrelational manner, the way the software generates music.

Conclusion

This paper has attempted to show aspects of the distinction between virtual and acoustic spaces in terms of loudspeaker use. Virtual space, with currently available technology, can be compared to film. Here a convincing visual illusion is created that incorporates three dimensions. The audience however, is unable to walk into the film and experience it as a three dimensional event. Like the screen in the cinema, the loudspeaker is the barrier to virtual space. There are currently developments in virtual reality that enable users to walk around in computer-animated spaces and it is conceivable that this might be possible in a sound landscape also. A complete solution to this problem may require the development of new technology that can project illusory sound sources into the listening space. For the time being it seems that the model in which loudspeakers are holes into a sound world outside that of the listener prevails.

This is not to suggest that the spatial limitations in relation to loudspeakers necessarily limit the creative possibilities for electronic music. However a classification of virtual spaces, which this paper attempts to outline, and an understanding of the perceptual relationship of the listener to these spaces, may be useful for the development of sophisticated compositional technique.

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Interactive sound works in public exhibition spaces: an artist's perspective

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An outline of recent research and installation works undertaken by Australian sound artist, Garth Paine.

Introduction

The focus of this paper is to introduce the work I have been doing over the last couple of years in the area of responsive environments. I have been exploring responsive environments within the scope of an artist, interested in using interactive sound to encourage a consideration of our relationship to our environment.

Context

The focus of my research over the last two years has been in the area of responsive environments. The area of interactive sound systems can be loosely categorised into three differing approaches:

- Gesture based interactive instruments
- Virtual Reality
- Responsive Environments

The differences in intent are marked.

- Gesture based interactive instruments can be seen as a development of the acoustic instrument, performance paradigm. These include a number of interfaces developed to mimic existing instruments, (such as Leonello Tarabella's piano interface), or to take existing instrument gestures and apply them to new synthesised output (see the work of Tau Tanaka and Sensor Band amongst others).
- Virtual reality (VR) tends to be focused on the creation of an immersive audio and visual environment. These environments are predetermined. They contain a limited number of paths or scene transitions the user can traverse. They do not provide a realtime generation engine that provides audio and visual stimulation unique to each user. VR pieces require the user to don sophisticated hardware interfaces that mediate ones presence and interactive potentials. They require a degree of familiarity with the hardware, and a learning of the interface. Virtual reality is illustrated in the work of Brenda Laurel, Christa Sommere and Laurent Mignonneau amongst others.
- Responsive Environments can be based on realtime synthesis engines, obviating the need for pre-made content. They do not require the wearing of hardware to interface with the system, and as such provide a high degree of flexibility, and encourage individual movement patterns. The freedom of interaction and the ability to produce outcomes in realtime, make the responsive environment unique in its ability to generate truly individual outcomes. Responsive Environments are illustrated in the work of Myron Krueger amongst others.

My work has been based in the area of responsive environments, because my research goals have been to explore forms of interaction that contain as few predetermined factors as possible. I have focused on systems that directly reflect the uniqueness of individual participants' input.

Artistic Intention

As an artist my interest lies in reflecting upon the human condition. There are many aspects of our lives, and many facets of international relationships between differing cultures, governments and philosophical persuasions that we simply do not take the time to consider in our day to day lives. Addressing these issues on some level however is vitally important to the continuing co-existence of the many disparate parties that form the global community, and integral to the further development of understanding and insight into the relevance of differences within the global community.

I see art as the perfect platform for the consideration and expression of these issues. Visual artists, writers, dancers and musicians have explored these issues for centuries with profound results. One only has to look at the prestige attached to the great cultural institutions of the world to see that the product of this artistic endeavour has communal value. One might argue that the value attached to these institutions is purely financial, that the value placed upon the works they contain is a product of contrary economic principles - of course the economics of historical value play a part in their financial value. It would however, be far too cynical to contribute their communal worth solely to the financial market place. If that were the case, the works would be housed in private environments for private enjoyment, not in public institutions, accessible by all.

If we agree that these works have a communal worth, we in turn agree that artistic endeavour is of value to society. As I have pointed out, this case is already proved for the traditional artforms, writing, music, dance and visual arts (painting and sculpture).

One of the interesting questions is:

- How do we develop an understanding of the way in which new artforms fit into this theory?

The explosion of computer based technologies in the last forty years has resulted in many new forms of expression. To some extent, the reduction of all material to a binary base of zeros and ones dissolves the borders between the artforms. In fact a software algorithm that is used to process sound may also be applied to visual material, although with dramatically different outcomes.

This reduction of all materials to binary soup has both advantages and disadvantages:

The advantage is that the synthesis of aural and visual events is possible. The artists can write an algorithm that produces a sequence of zeros and ones that generates an outcome descriptive of something that occurs in the natural world, or for that matter, something that has never, and may never possibly occur in the natural world.

For me though, the excitement of this ability to turn anything into a digital soup, is the potential that lies in the treatment of material gathered in the natural world. The digital domain facilitates the subversion, expansion, dissection, and general exploration of the “real world” material. It allows the artists to search the, until now, hidden aspects of the material for new expressions. A sound for instance has an outer face, that we may be familiar with. However, if one starts to dissect the sound, it exhibits many layers of interacting vibrations that can be teased out. Once the artist has control of these individual aspects of the sounds character, it becomes possible to accentuate the hidden characteristics, or to peel away the outer facade like an onion skin, thereby revealing finer and finer nuances.

The computer-based arts have been stymied by the complexity of the programming required to generate even the most simplistic and subsequently relatively boring image or sound. Nature in its wisdom in infinitely complex and variable. Naturally occurring sounds are often made up of extremely complex

combinations of partials which vary over time in elaborate ways. These variations are determined by the many environmental factors present in any momentary event.

Fortunately technological developments are currently so rapid that the speed of desktop computers has accelerated exponentially - the Apple Macintosh G4, presents specifications that only a few years ago were the domain of the super computer. The Apple Macintosh G4 or the Symbolic Sound Capybara 320 systems both provide the ability to generate audio in realtime that reflects the complexity inherent in nature.

This relatively affordable computing power has allowed artists like myself to move from *off-line* image and sound generation to realtime synthesis. Whilst this may not appear overly exciting, it has precipitated a much more profound exploration of the genre of interactivity.

Real time computation has allowed interactive arts to create realtime causal feedback loops.

The exploration of the cybernetic paradigm of feedback loops has been the focus of my installation works **Moments of a Quiet Mind**(1996), **Ghost in the Machine** (1997), and with the development of realtime synthesis engines, in my recent works **MAP1** (1998) and **MAP2** (1999/2000) and **REEDS** (2000).

MAP1 is an interactive sound installation commissioned in 1998 by the Next Wave Festival, Melbourne, Australia**. MAP1** uses realtime granular synthesis of live audio input as the interactive sound response. The sound is gathered using a microphone in the installation space. The granulation process (generated in SuperCollider on a Macintosh computer) is controlled by gesture analysis of people within the installation. The gesture analysis is carried out using the Very Nervous System (VNS - developed by David Rokeby), and a small black and white video camera mounted in the roof of the gallery. A MAX patch converts the variations in light intensity per pixel per video frame reported by the VNS into MIDI continuous Controllers (MCC). These MCC values are sent to the SuperCollider patch using the OMS IAC bus.

The granulation process uses a cycling audio buffer as its source. The audio buffer is overwritten when sounds made by its participants exceed a certain threshold. The audio buffer is overwritten for the period the sound remains above the threshold. This provides the opportunity to work with sounds entered by other participants (past or present), or to enter one's own sound sources

MAP1 therefore provided an environment in which the sounds made within it are resynthesised in response to the quality of the movement and behaviour patterns sensed by the system. The possibility for unique system responses is multiplied by the gathering of individual's soundings, and the analysis of individual gestural motion as the source of the resynthesis. In these ways **MAP1** provided a tight relationship between the input of each participant, and the system response. No pre-made material is contained in this installation.

Pierre Boulez describes composition as a selection of notes derived from a finite predefined set.
Trevor Wishart (On Sonic Art) points out that contemporary composition, especially within the genre of electronic music, goes well beyond “a finite lattice and the related idea that permutational procedures are a valid way to proceed. . .”. Wishart proposes a “musical methodology developed for dealing with a continuum using the concept of transformation”.

The concept of a stream of constantly evolving sound is directly supported by the use of realtime sound synthesis. The ever evolving - sometimes audible, sometimes not - processes of data driven art follow the Wishart approach. The Boulez approach is more closely aligned with the commercially prevalent paradigm of interactivity as a response to a defined challenge with a prespecified finite outcome, such as the triggering of existent sound files.

Realtime synthesis provides a subtle but profound alternative. The use within the synthesis instrument of variables controlled directly by movement gestures provides a way of not simply creating a personalised

mix of existent sound samples - much like a DJ - but of creating a completely unique sound stream. The temporal form as well as the pitch/timbre and "orchestration" of the score are created in realtime by the user. In my opinion, this freedom of response creates an individualised outcome that so tightly reflects the actions of the user as to be both qualitatively and quantitatively superior to an installation utilising pre-made sound sample events.

An installation using existent sound samples can be made to reflect user input by varying the playback polyphony, sample choice or small amounts of pitch bend. These relatively course reflections of control input does not reflect small intricacies of movement in as symbiotic a manners as realtime timbrel, envelope or modulation variations.

MAP2 features the concept of stream driven interactivity within an exploration of pure synthesis. **MAP2** was commissioned by the *Staatliches Institut für Musikforschung* (SIM), Berlin, and was exhibited at the Musical Instrument Museum, Berlin (Dec 1999 - Jan 2000). **MAP2** was developed in collaboration with Dr Ioannas Zannos.

MAP2 is based on a video sensing approach (using the VNS), that divides both horizontal and vertical planes into numerous independent fields. The horizontal plane is divided into four independent zones, each consisting of thirty-two fields, reflecting four different synthesis instruments, active at different thresholds of activity, and played by positional information provided. The vertical plane is defined as a row of interactive fields at a little over head height. These vertical fields control the playing of a physically-modelled plucked string sound.

The performance of **MAP2** is controlled by data gathered using the VNS. The horizontal space in **MAP2** was divided into four sections to enable participants to determine their own distinct input when playing the installation with others. This development was a response to feedback from users of **MAP1** who sought to clearly identify their own contribution to the interactive sound environment. Each of the four quadrants of the **MAP2** installation were totally independent, being controlled by separate data streams and addressing separate synthesis processes.

The research developments illustrated in **MAP2** are

- the simultaneous sensing of multiple interactive subjects with independent control over separate synthesis processes
- the sensing of the physical space as a 3-dimensional cube, applying separate video analysis to both the vertical and horizontal planes.
- the provision of multiple asynchronous synthesis processes, each with multiple synthesis instruments responsive to varying thresholds of movement activity.

These changes in the structure of the installation provided a many fold increase in the complexity of possible interactive responses over the **MAP1** installation. **MAP1** provided a single synthesis process for all interactive input for the entire physical space.

MAP2 marked a progression towards multiple asynchronous feedback loops - one for each division of both the vertical and horizontal sensed spaces, and one division for each of the synthesis instruments available within each threshold range of each horizontal field.

MAP2 also provides more variation in sound aesthetic and "*orchestration*" by virtue of the greater number of synthesis algorithms available, and the ability of the system to allow 4 people to interact simultaneously and asynchronously.

Whilst the change in dynamic of sensed movement causes additional synthesis algorithms to become active, the position of the sensed body causes variation in resonant filters placed in the signal path just prior to the each of the 8 independent audio outputs. The audio signals were sent to the output assigned to

the loud speaker closest to the position of the sensed activity. This technique caused the sound output to track the interactive behaviour through the physical space, two speakers being allocated to each of the quadrants specified in the horizontal video sensing set up.

My most recent work **REEDS** was presented as part of the Melbourne International Festival of the Arts, Melbourne, Australia in November, December 2000.
REEDS extends the approach to complexity of interaction illustrated in **MAP2** by providing eight simultaneous but asynchronous data inputs. **REEDS** changes the focus of the past works by moving away from the human body as the central controller to the adoption of momentary weather conditions as the data source. **REEDS** uses two weather stations, both collecting the following data:

1. Wind Direction
2. Wind Speed
3. Solar Radiation
4. Temperature.

The data is transmitted back to a land base, where it is parsed and directed as MCCs to a SuperCollider patch, containing six independent instruments (two stereo and four mono), containing eight variables and creating eight channels of audio.

The use of momentary meteorological data allows the exploration of truly chaotic multifaceted patterns of interaction.

The weather conditions naturally scope the range of response. The chosen synthesis approach and the specifics of the incoming data mapping on to the synthesis variables generally establish the aesthetic.
The use of naturally occurring data patterns is a way of exploring both the concept of interactivity as a stream, and an attempt to discover techniques for more tightly linking the organic and human contexts with interactive systems.

The program notes I wrote for the **REEDS** project illustrate the central tenet of the work:

> *A weed, so easily crushed underfoot, can push its way up through a tarmac path, creating a sizeable fracture in what appears to us to be an impervious surface.*
>
> *One might postulate that if it could see the bigger picture, it might have decided to grow 2 feet to the left in the flower bed or the grass. There is clearly an analogy here to our own birth, which we seem to have little or no say in (depending on ones religious bent).*
>
> *It is exactly this chaotic behaviour of the natural world that informs the Reeds project. Whilst civilisation tries to harness or tame the chaotic in nature, or to explain it in terms of quantum theory and fractals, humanity cannot perceive a truly chaotic state. The forces of nature that dictate the growth of plant life fall into this category. It is not possible for us to predict with certainty the meteorological conditions from day to day, let alone year to year, and certainly not on the micro scale of the weed in the footpath. It is precisely these chaotic variations that are used in Reeds to conduct the sound score - to control and dictate the output of the real time synthesis process.*
>
> *Of course, the software design process predetermines the general structure and aesthetic of the sound, but the momentary output is unique. It is unlikely that the combination of wind speed, wind direction, solar radiation, and temperature that occur in this instance will be precisely replicated in any other moment. This chaotic variation is the very source of diversity, which I propose is the structure that creates such beauty in nature.*
>
> *Reeds uses the relatively static external facade of the sculptural form as a way of representing the paradox observed in organic plant life, where in contrast to the apparently static external*

face of the plant, is the hidden, dynamic activity of photosynthesis and nutrient gathering that keeps the plant alive.

The Reeds pod sculptures appearing as lifelike presences on the Ornamental Lake at the Royal Botanic Gardens Melbourne, support two remote weather stations. These gather wind speed, wind direction, temperature, and solar radiation data (the meteorological conditions, vital to the plants life processes). The data is transmitted back to a land-base where it is transformed into eight channels of musical sounds that are broadcast back out to the Reed pods. These sounds give a voice to the secret activity of the inner life processes of the plant.

The viscous and fluid aesthetic of the sound material is an attempt to capture something of both the dynamism of the life sustaining processes and the ever-changing, silken thread that is the presence of life, the life force itself. The fact that the sound material is generated on the basis of meteorological conditions is a way of drawing as tightly as possible the bond between the processes of nature and the processors of the Reeds installation. The sound material can not then be avoided, being the voice of the processes of nature.

Sound/music is in many ways a unique media, for it is not an external artefact. Sound literally penetrates the body. Furthermore, it is impossible to concretely tie composed sound or music to a representation of anything beyond a communication of emotional states and journeys.

As an artist my interest lies in exploring ways of contextualising digital art processes within the natural organic environment. I have little interest in the purely synthetic, that is the synthesis of sound or images from a wholly academic or theoretical viewpoint. I prefer instead, as is illustrated in the Reeds project, to take a fundamentally organic source as the basis for my sounds. In so doing, I hope that some quality of that organic material will permeate the work and thereby bring the synthetic output at least a small way towards the organic world, and therefore within the human context.

Conclusion

It seems appropriate to ask about the value of interactive new media art works. What do we take away from them? How do they enrich our understanding of the world? Do we continue to think about the experience afterwards, thereby developing a deeper appreciation of the ways in which that experience reflects upon our own lives? — As one does well after viewing a good film.

I don’t pretend that my own work has these outcomes, although, like many other New Media artists, I strive to create work that will facilitate these outcomes. I must also say that a new media art is no where near its zenith. There is much work to be done in developing a language that communicates clearly and is sufficiently varied to accommodate the many individual artists working in the medium.

So I ask myself if the experience of these works is simply one of mapping the development of the art form, and in turn the evolution of the technologies, or an unbridled expression of artistic intent. I think that we are lucky enough at this point in the development of New Media art to experience both, however there are still many works that I experience, even at prestigious festivals that I think communicate little more that a technical achievement.

If new media art is to be taken seriously as an artform that has the capacity to communicate something of the metaphysical, we need to lose the technology — the technology that makes the work possible — the hours/weeks/months of programming, the innovative technical development. These aspects of the work, which are often revered as great achievements, needs to be translucent — conspicuous by their absence. The visitor/user should be unaware and unconcerned with the technology creating the experience. They should, however, experience a symbiotic relationship with the work that permits a real sense of freedom of interaction, and an infinite scope for self-expression and exploration. This is my goal, a goal I hope you see illustrated in the description of the work above.

In summary, I feel the most rewarding outcomes in responsive/interactive environments continue to be achieved through the exploration of realtime sound and/or vision generation. The realtime synthesis process reflects small intricacies of individual interaction. The participant feels directly acknowledged by this direct reflection of individuality. This in turn encourages a deep level of commitment to the exploration of the installations potential. Although technology has developed in leaps and bounds in the last decade, I feel computing power is just now sufficient for realtime interactivity. The current state of technology is encouraging for the development of this kind of work, We are living at a time that encourages realtime data driven sound synthesis through fast computing and excellent software tools.
We must shift our focus from technical achievements to a user driven experience. The technology must become both infinitely variable, and invisible to the end user.

The development of virtual reality technologies has shown a distinct partiality to the visual. In my view, sound is a much more direct and affective stimuli.

If we can make sound more responsive to individual interaction intricacies, I am sure we can prove responsive sound environments to be a superior form of immersive experience.

Found sound

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A review with excerpts from 3 sound features broadcast by *The Listening Room,* the ABC's Radiophonic program that goes to air on Monday nights.

The pieces are:

Resava Cave…….by Arenje Jovanovic
Stone as a musical instrument. A composition using percussive sounds from rock formations in a cave beneath a Monastery in Croatia.

Wake for Tom……. by Jim McKee and Barney Jones
Speech as music and poetry. Stories from homeless people on the street.

Currawong……. by Moya Henderson
Found sounds at the edge of the forest expressed in quasi-classical musical form.

Introduction

I am completing a PhD thesis on the subject of The Listening Room, a program on ABC Classic FM Radio devoted to sound art; I am writing about its history, the part it has played in the development of sound art in Australia and in the development of Australian composers, and the place it occupies within the world culture of sound art broadcasting. Every week, for about 11 months of the year since 1988, The Listening Room program has gone to air on ABC Classic FM; I have selected the first 10 years of its existence for detailed study.

I am literally working within The Listening Room in that I have a working space, access to files and scripts, archival and current tapes and CDs, people to talk to, library, data base and so on.

Going back to the 1970s and 80s, as portable sound recording equipment became smaller and lighter, and improved in quality, so more sounds were found. Some members of The Listening Room team still speak of the thrill they experienced when first hearing the recorded sounds of tiny insects chirping and clicking. All pretty old hat now, but significant in the history of sound art. So it was that sounds that previously had not been available for use, you might say, as instruments became available as technology advanced.

In the course of my work at the ABC I have listened to many compositions using found sound and I have selected three of these which I have found particularly appealing. I find them engaging pieces, technically of broadcast quality and possessing a strong aesthetic content. Each is an example of a different kind of found sound. Each is an example of the skill of composition. Here are excerpts from them.

Resava Cave

Broadcast date: 29/2/93
Composer: Arenje Jovanovic
Producer: Arenje Jovanovic
Duration: 14 minutes.

Sounds for *Resava Cave* were recorded in a cave of that name in Serbia by Arenje Jovanovic and a group of his students and 2 musicians. It won the Prix Italia at the International Radio Festival of 1989, an award that Jovanovic had then won several times. It has been put to air by The Listening Room 3 times during the 10 year period under my review.

The Cave is some 15 m. into the cliff face below a 15th Century Monastery just outside Belgrade and contains the stalagmites and stalactites that form the instruments played by Jovanovic and his students. Using sticks and stones and hands the players generate a great variety of sounds, sounds that in most cases express their character across a range of frequencies, depending on how and where the formations are struck. Selected and edited by the composer, and combined with the human voice, these are the components of the composition. In his introduction to the feature Jovanovic describes it as one of the most important experiments in the development of his composition of sound art.

The various 'instruments' used produce tones of great beauty and character. The sound that is created when a stalagmite or stalactite in the cave is struck is a truly musical note with a clear fundamental, and harmonics that express the nature of the material. That the material is immediately recognizable as stone is, I believe, a significant contributer to the aura of *Resava Cave*. The sound conveys the essence of stone, its noble characteristics of, for example, strength, stability, endurance and timelessness. The sound it makes when struck becomes a musical metaphor for these qualities, and of course others. This must be one of the advantages of working with sounds from the natural world, as in the minds of listeners there already exists a perception of characteristics associated with a particular sound.

In structure the piece contains elements of conventional musical form, with crescendo and de-crescendo, approach and retire, movements across the soundscape, climaxes and silences. Sometimes the quiet episodes are filled with the soft throbbing of insects or sounds like the beating of a heart or the gentle breathing of sleep. At times the haunting voices of women float through the air, contrasting with the percussive sounds of sticks hitting solid resonant rock. As the piece progresses the simple lines of the beginning change slowly to complex layering and increasing sound levels, rising to a plateau. Then simplicity returns and the music comes to a placid ending. It is easy to imagine how the effect of *Resava Cave* comes from the improvised generation of sounds, by candlelight in an echoing cave many centuries old.

The significance of the feature I believe lies in the imaginative choice of a sound arena, and the beauty of the resulting composition; the choice of sounds and their arranging that bring to the imagination of the listener feelings of awe, a quality of history and immersion in the resonances of the cave.

Wake for Tom

Date to air: 18/4/94
Producer: Jim McKee
Composers: Jim McKee and Barney Jones
Duration: 28 minutes

At the time of composition Jim McKee and Barney Jones together ran the business Earwax Productions in San Francisco where they made their living by creating sound material for commercial purposes, their customers mainly small independent media outlets. They were well paid for their work but mostly they had to deal with current matters, in the way of a cartoonist, and meet short deadlines. *Wake for Tom* was different; it was funded by a small grant, way below the usual hourly commercial rate, its composition

stretching over a period of about 12 months. Their subject matter was just outside the door, on the street, the homeless alcoholics in the Tenderloin district of San Francisco.

In 1993 Jones and McKee won an Oscar for their sound editing of Cuppola's film Dracula. They had also made in 1987 what they call an 'Audiograph' titled *Pop songs from the Tenderloin; Wake for Tom*; it was their second Audiograph and was an entry in the 1993 Berlin Prix Futura.

To find out more about the history of *Wake for Tom*, Listening Room Producer Robyn Ravlich talked with the composers by phone from Sydney and this conversation became part of the broadcast Introduction of the piece. The composers recount how, by edited conversations with men and women of the street people, the story of living and dying on the street and in particular the death of Tom, is told by the people themselves. Their voices are the only voices heard, set over a running theme of piano music, plus other conventional instruments from time to time.

With no payment attached to the outcome and no deadline for its completion, *Wake for Tom* developed within a lengthy time-frame, with McKee and Jones being able to consider the results of what they refer to as their own compositional research. Although the composers had no real idea of what would happen as they worked on the recorded material, they had the object of turning Interview into Theatre, of making music from spoken words. In the phone conversation with Ravlich one of the composers says: 'There is so much music in what people say, there are tunes and rhythms to be turned into music — you may as well be poetical, make it flow as music'. They go on to describe 'throwing the voice about to where the music is', and in another part of the conversation they talk about laying the music into the speech structure.

What the composers set about doing was to find, in the mass of recorded material, voices that expressed in both sound and meaning what they wanted to communicate, and 'throw' these into the music of the piano. 'Throwing' was helped significantly by the use of digital mixing technique, which was apparently new to them at the time and took the place of bringing analogue tape tracks together. It was quicker and more accurate; there was no 'that'll be good enough', it could all be done so as to satisfy exactly the composer's creative intent.

There are no actors, no interviewers, no narrators in this Feature; the story is told by the street people. Through their words their world is described in little musical poems, each one telling about some part of it; waking, sleeping, begging, dying. The poetry of words comes through in expressions such as:

> Spare a little change?
>
> Let's see, Howard died in August.
> Carol died in September.
> Steve died in November.
> We're waiting for December.
>
> You got someplace where you can store these ashes for a while?
>
> You can't panhandle here.
>
> Spare a little change
> So I can stay drunk and unemployed.
> A dime for a bottle of wine?
> At least I'm honest.

Melded with sounds of the piano, the story emerges as a musical event, its own rhythm and melody so subtle as to be unremarkable, but with a tempo and shape consistent with the story.

So written, in what they describe as a 'song structure', and through their choice of the words spoken, the composers believe that they may have helped reveal some kind of essence in the existence of these people. They believe that perhaps in an accidental way their work communicates with 'truth' or at least expresses with 'objectivity', the life of the street people. This goes a step further than the suggestion that found sounds go to make an interesting or engaging musical composition. The composers are suggesting that by adding their minds and skills to sounds that are part of a culture they reveal or communicate an essence or spirit of that culture not usually perceptible..they have been instrumental in discovering and imparting a truth.

Currawong

Broadcast date:8/8/88
Composer: Moya Henderson
Duration: 22 minutes.
Commissioned by WDR (West German Radio) as a co-production with the ABC.

Australian composer Moya Henderson was commissioned by Klaus Schoning of West German Radio to write a piece for co-production with the Australian Broadcasting Corporation for broadcast by Schoning's Studio Akustische Kunst. Originally named *Currawong — a symphony of birds*, the sounds were recorded by Sound Engineer Phillip Ulman at Barrington Tops, a wilderness area a few hours drive north of Sydney, at dawn as the birds were feeding at the garbage containers of the local Guest House. Later, Ulman recorded the calls of Bellbirds and other varieties in nearby trees, thus presenting the composer with a range of bird song and environments from which to make her creative selection. Henderson writes that her 'job as composer was to highlight the drama and humour of these early morning feeding frenzies, and to enhance the listener's awareness of the extraordinary polyphony that fills the dawn skies over the Barrington Tops Guest House'.

Moya Henderson writes that broad symphonic form inspired the way in which the sounds were put together and this aspect was examined further by Sally Macarthur of the University of Western Sydney in her doctoral thesis. In advancing an argument supporting this statement Dr Macarthur refers to her aural analysis of the piece (there is no score) and derives from that four sound groups, viz Introduction, Exposition, Development and Recapitulation, each with its own internal elements capable of being identified in broad classical musical terminology, i.e. first subject, second subject and so on. Macarthur proposes that in selecting from the recorded material presented to her by the sound engineer, Henderson followed a path suggested by her training in Classical form. Another viewpoint on the matter of compositional structure is that in setting out to `highlight the drama and humour of these early morning feeding frenzies and to enhance the listener's awareness of the extraordinary polyphony that fills the dawn skies' etc. Henderson was concerned with communicating. Her selection of the sounds presented to her and the sequence of their `appearance' springs from her creative objectives. Whatever form there is flows from that. Although Henderson believes that, in its broadest sense, the symphony has inspired the way in which all the sounds have been put together, she writes in a letter on ABC files: `In the end though, the content dictates the form'.

Currawong is the sounds of birds and of their activities: the teaching of their young - claiming of territory - early morning awakening - in human terms, what sounds like gossiping, and feeding from garbage containers. All sounds speak of the open air, and the sounds accompanying the feeding, such as the metallic resonance of garbage tins and lids and the scraping of talons on corrugated iron, invoke a powerful immediacy. The distant calling of birds in the forest behind the feeding area mark out the depth of the sound-stage and go towards defining the total space.
But as much as our understanding and appreciation of *Currawong* may be based on reflections of human behaviour, the listener is not lured into the anthropomorphic trap. Several times during the piece songs and calls fade and we hear, and almost feel, the powerful beating of wings and the rush of air as a bird lands near a microphone. This is an unmistakable message that these birds live in an element that we cannot enter and it does not befit us to consider these sounds, while the birds feed on our scraps, as being produced by some sort of sub-human circus.

Studio Akustische Kunst at West German Radio is an important player in the world of radio sound art and Klaus Schoning of WDR is highly respected by his peers in many countries. That *Currawong* was jointly commissioned by WDR and the ABC is an important event giving a clear indication of the esteem in which the ABC and Moya Henderson are held on the International scene. WDR has broadcast it twice; Schoning refers to it in a personal comment as 'a beautiful piece about these wonderful birds'.

As a matter of interest, Henderson had an early association with Germany, as a scholar in the German Academic Exchange Service. She studied with Maurice Kagel at the Musikhocschule of Cologne from 1974 to 1976.

References

Macarthur, Sally Ann. (1997) *Feminist aesthetics in music: politics and practices in Australia*, Thesis for degree of Doctor of Philosophy, Department of Music, University of Sydney. pp.361-367.

Schoning, Klaus. (1996). *Acustica International Lydrejse*. Pages un-numbered. Section in English. Museet for Samtidskunst, Roskilde, Denmark

Transposed Hexanies: Just intonation synthesis studies

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Just intonation tuning systems that have no tonal centres provide the starting point for this discussion on tuning associated with a series of works entitled *Transposed Hexanies*. The composer describes the origins of the hexany, the hexanies used in this these works, their transformation and extension through pitch multiplication and the connections between rhythms and tunings used in these works. The lecture will demonstrate how transposition extends the scope of the hexany. The transposition algorithm will be explained together with its ramifications for live performance using hand-held instruments capable of performing in tunings other than twelve equal divisions of the octave.

1 Motivation : understanding Wilson's Combination Product Sets

Over the past few years, I have been fascinated by the theory of Combination Product Sets, first proposed by Erv Wilson in his article "D'alessandro, Like a Hurricane" in Xenharmonikôn 12 (Wilson 1989) and which he gave me in 1998. I am particularly interested in the unique complementary relationship between major and minor tonality inherent in his theory. I have explored this relationship in a series of eleven studies in just intonation called *Transposed Hexanies*, which I began in May 2000. The first of these studies was premiered at ACMA 2000, Queensland University of Technology, Kelvin Grove on Jul 5th 2000; a further ten studies *Transposed Hexanies* No.2 –No.11 were premiered at MicroFest 2001, Pomona College, Claremont, California on 6th April 2001.

1.1 Combination Product Sets — the hidden tonic

Combination Product Sets are just intonation scales formed when combinations of a set of harmonics are multiplied together. One of the simplest kind is the hexany, a 6-note scale generated by multiplying all combinations of pairs of harmonics from a set of four harmonics. For example, by combining pairs of products the first four odd harmonics 1, 3, 5 and 7, as shown in Table 1a, the pitch products shown in Table 1b are produced. In just intonation, these are expressed as fractions with each pitch (numerator) relative to a tonic (denominator) as shown in Table 1c. Note that the tonic is only implied and is not present in the scale. Pitches have been arranged in ascending order.

My initial observations of the hexany were focused on just major and just minor triads found in the above hexany. In Table 1c, pitches in the odd numbered columns form the just major triad, while pitches in the even numbered columns form the just minor triad.

These complementary major and minor triads are found on diametrically opposed sides of Wilson's geometric representation of the hexany as shown in Figure 1. Wilson's hexany is represented as two super-imposed triangles, one representing tetrads formed by the products of each harmonic, the other its complement formed by the products of each sub-harmonic. Unlike Partch's diamond, which also organises chords around the harmonic and sub-harmonic series with the tonic explicitly included, Wilson's combination product set is "a structure which could produce tonal progression without a tonal centre" (Rapoport 1994). The novelty of the combination product set structure is that the tonic is absent from the scale altogether as represented by the hole at the geometric centre, yet at the same time, it is tonal because of the complementary harmonic relationships it creates.

I wanted to know what sense my ears would make of complementary harmonic relationships in various other hexanies generated using a different set of harmonics. I made a 1 3 5 7 hexany using *Scala*, tuning analysis/editor/librarian software written by Manuel Op de Coul, and Csound, computer music synthesis software written by Barry Vercoe and others. This hexany is heard in the first audio example; the odd numbered notes of the hexany -- ie. a 1st inversion major triad -- are played first, followed by the even numbered notes – a root position minor triad (audio example 1). I played with the Csound code to find out how these major and minor triads sounded in relation to other triadic combinations found within the hexany (audio examples 2a and 2b). I tried hexanies generated using other primes besides 3 5 and 7 but went no further than the prime 11. I restricted myself to hexanies produced from combinations of harmonic generators shown in Table 2. Pitches of the five hexanies are shown in Table 3 (audio example 3).

1.2 Transposition algorithm : the tonic reappears

I also wanted to know what happens if pitches of the hexany are transposed using the intervals of the same hexany. I modified the Csound source code to allow transposition. The Csound Orchestra is shown in Example 1a, the Score in Example 1b.

Transposition introduces interesting possibilities for voice leading, as demonstrated in the following passage from the first study written in May 2000 (audio example 4).

I wrote a scala command file to find out what actual pitches are produced by transposing a hexany. I discovered that transposition replicates some pitches of the original hexany and produces unique pitches ie. pitches that are not found in the original hexany or pitches that are not duplicated in other transpositions. Each hexany when transposed only by itself, produces 19 unique pitches. The total number of unique pitches in the five 6-note scales is 9. The total produced through transposition is 69, a ratio of more than 7 : 1.

Transposing the hexany does more than create additional pitches. By virtue of the hexany's limited set of harmonic relationships, transposition organises a large collection of pitches into smaller manageable groups of pitches. All pitches available through transposition in Hexany 1 are shown in Table 4a; Hexany 2 (4b); Hexany 3 (4c); Hexany 4 (4d) and Hexany 5 (4e). One intriguing thing about these transpositions is that the tonic, or 1/1, which is hidden in the original hexany, re-emerges an octave higher in the first four of these tables. In the following examples all pitches for each transposed hexany are played as complementary major and minor triads (audio examples 5a).

Transposition is implemented in Csound using pitch multiplication. Two linear factors are multiplied, each factor selected from two parameter fields in the Csound score. One parameter field, p5, selects a pitch from the original hexany; the other, p6, selects a transposition interval. In this way, I was able to work with the harmonic features of a single hexany and, quite separately, experiment with its various transpositions.

The Csound implementation of transposition made it a simple matter to transpose the pitches of one hexany, p7, using the intervals of another hexany, p8. This technique, which I will refer to as cross transposition, became the basis of the next ten studies in the collection, *Transposed Hexanies No.2–No.11*.

2 Transposed Hexanies No.2-No.11 (2001)

The ten studies *Transposed Hexanies No.2–No.11* employ all cross-transpositions of five hexanies. Each study is based on one of five hexanies shown in tables 1 and 2. In each study, pitches of one hexany are transposed by intervals of another. A different cross-transposition is used in each study. The ten possible combinations are shown in Table 5.

In each of the ten studies, transposition does not occur continuously throughout the entire study. At some point, each hexany is heard untransposed; elsewhere in the same study, it is transposed at my discretion. This is done in order to modify the harmonic context at various places in each study where I thought some unexpected musical change was warranted. The effect of transposing one hexany with another is analogous to the leaching technique used by artists who work with water colours; new hues of colour are created when a second layer of colour is added while the first layer is still wet.

2.1 Rhythmic templates

In each of the ten studies, a number of rhythmic templates were created using standard music notation software, Finale. I created rhythmic templates using ratios I found in the hexany. Each template is organised in four layers, each layer assigned to a dedicated stave. Each stave corresponds to one of the rhythmic generators used in each study. Each template is saved as a MIDI file which is converted into a Csound score file using MIDI2CS, public domain software developed by Rüdiger Borrmann. The Finale score was never intended as a challenge to a virtuoso performer. It is used solely as a visual means to organise six pitches played as complex rhythms, and enter it into Csound. Once converted to Csound score, I arbitrarily selected the tessitura, transposition and hexany numbers by altering values in parameter fields p5, p6, p7 and p8 of the Csound score. Example 2 shows Hexany 5 in standard notation.

2.2 Gray code sequences

In each of the studies, Gray code is used to create an orderly sequence of states through every combination of rhythmic generator. Gray code creates a transition from one rhythmic template to another whereby only one layer enters or exits at any one time, as shown in Tables 6a-e. Gray code combines these strands of rhythmic information in a single seamless transition from one state to another without any combination being repeated. Each of the four rhythmic templates is heard alone once; each template is heard once in combination with one other template; each template is heard once in combination with two other templates; all templates are heard once together (musical example 6a).

3 Conclusion

Working with the hexany invites a further development of the boolean harmonic filter algorithm I created for the MIDI Tool Box (Schiemer 2000). This algorithm was previously applied in the context of 12-tone equal temperament. It was based on the premise that MIDI note numbers are really two interleaved sets of whole tone scales, one containing odd numbered notes, the other containing even numbered notes. In a just intonation scale like the hexany, where odd numbered notes define major tonality and even numbered notes define minor tonality, the boolean filter algorithm offers a computationally efficient way of manipulating harmonic fields in just intonation.

In its ideal state *Transposed Hexanies* will ultimately be a live perfomance work. At the very least, these studies, implemented in Csound, will provide the basis for a new live performance work. The current realisation is in mono. Some spatialisation will be added in a future implementation. The timbre is currently limited to Csound's pluck instrument. Future enhancements will involve plucked string portamento to give the quality of phrasing created by players of the Vietnamese monochord, known as *dan bau* or the quality of *gamaka* found in Carnatic vocal music.

Ideally, I look forward to a performance of the work as part of a new development project involving battery-operated hand-held instruments. The project will involve performance using current wireless technology based on Java MicroEdition and is based around the Motorola MCore microcontroller. This technology opens possibilities for performing music in ways that are no longer constrained by the current desktop computing paradigm.

Above all, the economy of the transposition technique developed in Transposed Hexanies makes it appropriate in mobile hand-held computing environment. Because it is implemented in a way that combines table look-up with arithmetic performed on-the-fly, it can easily be applied within the hardware

constraints of J2ME technology. Squeezing more out of less is not just a mindless musical pursuit. It is also a way of squeezing more data into small memory address spaces associated with K-Virtual Machine environments. Ultimately the technique could be useful in developing a new generation of mobile hand-held musical instrument capable of performing music tuned in just intonation.

References

Wilson, Ervin 1989 “D’Alessandro, Like a Hurricane” Xenharmonikôn 12 pp. 1-38

Rapoport, Paul 1994 “Just Shape Nothing Centra” Musicworks 60 pp. 42-49

Burt, Warren 1996 "An Interesting Group of Combination-Product Sets Produces Some Very Nice Dissonances" *1/1* 9 [1] p. 1

Schiemer, Greg 2000 “Boolean logic as a harmonic filter” Interfaces Proceedings of Australian Computer Music Conference 2000 pp. 59-67.

(1a)	**5 x 7**	**1 x 5**	**3 x 7**	**1 x 3**	**1 x 7**	**3 x 5**
(1b)	**35**	**5**	**21**	**3**	**7**	**15**
(1c)	**35/32**	**5/4**	**21/16**	**3/2**	**7/4**	**15/8**

Table 1 Derivation of pitches in a hexany

Hexany	Generators			
No. 1	**1**	**3**	**5**	**7**
No. 2	**1**	**3**	**5**	**11**
No. 3	**1**	**3**	**7**	**11**
No. 4	**1**	**5**	**7**	**11**
No. 5	**3**	**5**	**7**	**11**

Table 2 Generators used in the five hexanies

HEXANY 1	**Generators: 1 3 5 7**				**(audio example 3a)**
1: 35/32 septimal neutral second			**5**	**7**
2: 5/4 major third	**1**		**5**	
3: 21/16 narrow fourth		**3**		**7**
4: 3/2 perfect fifth	**1**	**3**		
5: 7/4 harmonic seventh	**1**			**7**
6: 15/8 classic major seventh		**3**	**5**	

HEXANY 2	**Generators: 1 3 5 11**				**(audio example 3b)**
1: 33/32 undecimal comma		**3**		**11**
2: 5/4 major third	**1**		**5**	
3: 11/8 undecimal semi-augmented fourth	**1**			**11**
4: 3/2 perfect fifth	**1**	**3**		
5: 55/32			**5**	**11**
6: 15/8 classic major seventh		**3**	**5**	

HEXANY 3	**Generators: 1 3 7 11**				**(audio example 3c)**
1: 33/32 undecimal comma		**3**		**11**
2: 77/64			**7**	**11**
3: 21/16 narrow fourth		**3**	**7**	
4: 11/8 undecimal semi-augmented fourth	**1**			**11**
5: 3/2 perfect fifth	**1**	**3**		
6: 7/4 harmonic seventh	**1**		**7**	

HEXANY 4	**Generators: 1 5 7 11**				**(audio example 3d)**
1: 35/32 septimal neutral second		**5**	**7**	
2: 77/64			**7**	**11**
3: 5/4 major third	**1**	**5**		
4: 11/8 undecimal semi-augmented fourth	**1**			**11**
5: 55/32		**5**		**11**
6: 7/4 harmonic seventh	**1**		**7**	

HEXANY 5	**Generators: 3 5 7 11**				**(audio example 3e)**
1: 33/32 undecimal comma	**3**			**11**
2: 35/32 septimal neutral second		**5**	**7**	
3: 77/64			**7**	**11**
4: 21/16 narrow fourth	**3**		**7**	
5: 55/32		**5**		**11**
6: 15/8 classic major seventh	**3**	**5**		

Table 3 Pitches of the five hexanies

```
              instr 1

              ihexany    =   p5           ;   select hexany pitch value
              itranspose =   p6           ; select hexany transposition
              ifrq1  cps2pch ihexany,    -(p7)
              ifrq2  cps2pch itranspose, -(p8)
              ifrq       =   ifrq1 * ifrq2
asig          pluck  ampdb(p4), ifrq, ifrq, 0, 1
              aright     =   asig
              aleft      =   asig
              outs aright*(givol*1.00), aleft*(givol*1.00)
              endin
```

Example 1a

```
; ------------------------ Transposed.Hexanies.SCO -------------------

f1 0 8 -2  1.00000000  1.09375000  1.25000000  1.31250000  1.50000000  1.75000000
1.87500000
f2 0 8 -2  1.00000000  1.03125000  1.25000000  1.37500000  1.50000000  1.71875000
1.87500000
f3 0 8 -2  1.00000000  1.03125000  1.20312500  1.31250000  1.37500000  1.50000000
1.75000000
f4 0 8 -2  1.00000000  1.09375000  1.20312500  1.25000000  1.37500000  1.71875000
1.75000000
f5 0 8 -2  1.00000000  1.03125000  1.09375000  1.20312500  1.31250000  1.71875000
1.87500000
; tempo
t 0 15.0000


; p1 = instr
; p2 = start
; p3 = duration
; p4 = dB
; p5 = pitch
; p6 = transposition (nb. octave must = 0)
; p7 = hexany N (pitch)
; p8 = hexany M (transposition)


;p1    p2       p3      p4      p5      p6      p7     p8

i01    0.00000  1.1250  80      8.01    0.01    1      1    ;  D3 ch01
i01    0.00000  1.1250  80      8.03    .       .      .    ;  F3 ch01
i01    0.00000  1.1250  80      8.05    .       .      .    ;  A3 ch01

i01    1.12500  1.0000  79      8.01    .       .      .    ;  D3 ch01
i01    1.12500  1.0000  79      8.03    .       .      .    ;  F3 ch01
i01    1.12500  1.0000  79      8.05    .       .      .    ;  A3 ch01

i01    2.12500  1.0000  78      8.02    .       .      .    ;  E3 ch01
i01    2.12500  1.0000  78      8.04    .       .      .    ;  G3 ch01
i01    2.12500  1.0000  78      8.06    .       .      .    ;  B3 ch01
etc.
```

Example 1b

	Original	Trans. 1	Trans. 2	Trans. 3	Trans. 4	Trans. 5
1:	**35/32**	**5/4**	**21/16**	**3/2**	**7/4**	**15/8**
2:	**5/4**	**10/7**	**3/2**	**12/7**	**2/1**	**15/7**
3:	**21/16**	**3/2**	**63/40**	**9/5**	**21/10**	**9/4**
4:	**3/2**	**12/7**	**9/5**	**72/35**	**12/5**	**18/7**
5:	**7/4**	**2/1**	**21/10**	**12/5**	**14/5**	**3/1**
6:	**15/8**	**15/7**	**9/4**	**18/7**	**3/1**	**45/14**

Table 4a Hexany 1

	Original	Trans. 1	Trans. 2	Trans. 3	Trans. 4	Trans. 5
1:	**33/32**	**5/4**	**11/8**	**3/2**	**55/32**	**15/8**
2:	**5/4**	**50/33**	**5/3**	**20/11**	**25/12**	**25/11**
3:	**11/8**	**5/3**	**11/6**	**2/1**	**55/24**	**5/2**
4:	**3/2**	**20/11**	**2/1**	**24/11**	**5/2**	**30/11**
5:	**55/32**	**25/12**	**55/24**	**5/2**	**275/96**	**25/8**
6:	**15/8**	**25/11**	**5/2**	**30/11**	**25/8**	**75/22**

Table 4b Hexany 2

	Original	Trans. 1	Trans. 2	Trans. 3	Trans. 4	Trans. 5
1:	**33/32**	**77/64**	**21/16**	**11/8**	**3/2**	**7/4**
2:	**77/64**	**539/384**	**49/32**	**77/48**	**7/4**	**49/24**
3:	**21/16**	**49/32**	**147/88**	**7/4**	**21/11**	**49/22**
4:	**11/8**	**77/48**	**7/4**	**11/6**	**2/1**	**7/3**
5:	**3/2**	**7/4**	**21/11**	**2/1**	**24/11**	**28/11**
6:	**7/4**	**49/24**	**49/22**	**7/3**	**28/11**	**98/33**

Table 4c Hexany 3

	Original	Trans. 1	Trans. 2	Trans. 3	Trans. 4	Trans. 5
1:	**35/32**	**77/64**	**5/4**	**11/8**	**55/32**	**7/4**
2:	**77/64**	**847/640**	**11/8**	**121/80**	**121/64**	**77/40**
3:	**5/4**	**11/8**	**10/7**	**11/7**	**55/28**	**2/1**
4:	**11/8**	**121/80**	**11/7**	**121/70**	**121/56**	**11/5**
5:	**55/32**	**121/64**	**55/28**	**121/56**	**605/224**	**11/4**
6:	**7/4**	**77/40**	**2/1**	**11/5**	**11/4**	**14/5**

Table 4d Hexany 4

	Original	Trans. 1	Trans. 2	Trans. 3	Trans. 4	Trans. 5
1:	**33/32**	**35/32**	**77/64**	**21/16**	**55/32**	**15/8**
2:	**35/32**	**1225/1056**	**245/192**	**245/176**	**175/96**	**175/88**
3:	**77/64**	**245/192**	**539/384**	**49/32**	**385/192**	**35/16**
4:	**21/16**	**245/176**	**49/32**	**147/88**	**35/16**	**105/44**
5:	**55/32**	**175/96**	**385/192**	**35/16**	**275/96**	**25/8**
6:	**15/8**	**175/88**	**35/16**	**105/44**	**25/8**	**75/22**

Table 4e Hexany 5

1

Transposed Hexany	2	3	4	5	6	7	8	9	10	11
Pitch Hexany	**1**	**4**	**2**	**3**	**5**	**1**	**4**	**2**	**3**	**5**
Transposition Hexany	**3**	**1**	**3**	**5**	**2**	**2**	**5**	**4**	**4**	**1**

Table 5 Cross-transpositions in *Transposed Hexanies No.2-No.11*

1		1	1	1	1					1	1	1	1		
3				3	3	3	3	3	3	3	3				
5								5	5	5	5	5	5	5	5
7	7	7			7	7			7	7			7	7	

Table 6a Gray Code 1

3		3	3	3	3					3	3	3	3		
5				5	5	5	5	5	5	5	5				
1								1	1	1	1	1	1	1	1
11	11	11			11	11			11	11			11	11	

Table 6b Gray Code 2

7		7	7	7	7					7	7	7	7		
1				1	1	1	1	1	1	1	1				
11								11	11	11	11	11	11	11	11
3	3	3			3	3			3	3			3	3	

Table 6c Gray Code 3

5		5	5	5	5					5	5	5	5		
11				11	11	11	11	11	11	11	11				
7								7	7	7	7	7	7	7	7
1	1	1			1	1			1	1			1	1	

Table 6d Gray Code 4

11		11	11	11	11					11	11	11	11		
7				7	7	7	7	7	7	7	7				
3								3	3	3	3	3	3	3	3
5	5	5			5	5			5	5			5	5	

Table 6e Gray Code 5

Figure 1 Wilson's geometric represenation of the hexany

Example 2 Hexany 5 in standard notation

Inframedia Audio

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Abstract

This paper examines local activity in non-academic electronic music, proposing that this work can be analysed as, among other things, a form of media art practice. It is proposed that an engagement with audio media as such is a central feature of this work; specifically, that this work can be understood as an attempt to hear, and experience, audio media substrates. The notion of inframedia is proposed, standing for media practices which operate in this way, below the level of content. Continuities are demonstrated between this audio inframedia practice and visual and audiovisual inframedia forms.

LEMu: Live Electronic Music Generator

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This paper examines the software instrument "LEMu", which is being developed by the authors at the Queensland University of Technology. In this paper we will answer the questions: "What is LEMu?", "What is it designed to do?", "How does it work?", "What music does it produce?" and "What are some future developments of LEMu?". Many types of intelligent instruments are emerging from both the academic and commercial worlds. For example there is the interactive algorithmic composition *AutoGam*, developed by Thierry Bachman and the Korg *KARMA* keyboard, developed by Stephen Kay. Presently the commercial dance scene lacks (to some extent) real time performance, creation and improvisation. LEMu encourages DJ's and electronic musicians to explore these concepts within the context of the dance music scene.

What is LEMu?

LEMu is a software instrument that works with MIDI events. It has the ability to create musical phrases and manipulate them in real time. It is predisposed to creating music for many underground and commercial genres of electronic music such as techno, house, trance, breakbeat and electro.

As LEMu works within the MIDI realm, a lot of the flexibility and musicality of the instrument depends on what type of synthesizer is creating the sounds. In our own MIDI hardware setup, we use the Alesis DM5 drum machine, the Roland JP8080 analogue modelling synthesizer and the Kurzweil K2000, with sounds that we have programmed in ourselves. This setup along with our musical understanding defines our own unique LEMu sound.

LEMu was programmed in the jMusic environment that is an extension of the Java programming language. jMusic was developed by Andrew Brown and Andrew Sorensen and was introduced to ACMA at Interfaces ACMC 2000 (Sorensen & Brown 2000).

What is it designed to do?

LEMu is a fusion of two major electronic music concepts – the academic and the commercial. As an academic concept it was created to more fully explore various interesting ideas concerning computer music. These are: realtime jMusic, aleatoric composition and mathematic manipulation of sequences. In respect to the commercial possibilities of the software, LEMu offers techno/dance music artists and producers, a new live performance and music creation system. This system grants the user many spontaneous performance abilities, some of which are similar to standard DJ techniques (layering of tracks, cross fading, slicing). Important concepts for this goal were: interaction, useable interface and design, storage and playback of tracks/patterns. In time, as LEMu evolves, it may gain support in the club scene where it will exist as an alternative medium for live performance of electronic music.

How does it work?

On opening the program the user is presented with the master interface.

From the master interface, the user can create up to sixteen sequence modules and synchronise the timing of the program (the speed may need adjustment on various different platforms). During playback, the master interface has general controls for the manipulation of the music such as tempo and master transposition. It also allows for the saving and storage of MIDI files (capturing a live performance). There are two types of sequencing modules – rhythm and pitch.

Upon creation the sequence modules are assigned to a MIDI channel. MIDI data from each module is compiled into a beat long score that is played using the QuickTime utility object from jMusic. QuickTime then routes the data to external MIDI hardware.

Pitch module

This module is concerned with the creation and manipulation of musical phrases. Universal module parameters include sound selection, pattern length, time signature, note density, mode, pitch set, phase offset, transpose, rhythmic shift, dynamic and mute state. These are controlled by the user and have values that are independent of the other modules. Phase offset offsets the start time of the phrase by semiquaver increments.

Rhythm module

The rhythm module starts with a preset pattern, which is looped. User can control sound selection and drum rhythms. We have defined eight parts to a drum module: kick, snare, high hats open, high hats closed, clap, cymbal, aux1, aux2. The preset pattern has a relevant sequence for each of these parts, the style menu option dictating the patterns necessary for each part. Each of these parts can be assigned to a MIDI note number so really the kick part could in fact be playing any sound if the user wishes to set it up in such a way.
Each part has a rhythmic shift slider and a mute button. The shift slider creates a polyrhythmic version of the part that is affected. It defaults to a value of four, signifying four beats to a bar, and can move from 1 to 8.

Capture

The capture function in the master interface is used to merge all the beat long scores into a single score that is then converted to a standard MIDI file.

What music does it produce?

The music created with LEMu has its own trademark. By this we mean that there are a few compositional ideas that are generally consistent from one song to the next. These are: layering, repetition, polyrhythms and the preset beats that are derived from the electronic dance music genre. These are the functions that LEMu performs most easily, thus defining the style of the instrument.

Future development

Most of the features that we have developed in LEMu will be refined and many new features are yet to be added. As there is currently no physical controller input into the software, effective manipulation is limited by mouse and keyboard operation. We have attempted to make the software interface more usable, which is necessary if LEMu is to be run independently. Some of the ideas we are yet to implement are:

interaction between parts, a more intelligent approach to the creation of musical phrases, a variety of algorithms to select from for the shift sliders (of which the current "polyrhythmic algorithm" will be one) and a chord/arpeggiator module.

In light of the release of Mac OSX we will be looking to develop LEMu using Java 1.3. The most obvious reason for us doing this would be that 1.3 includes the JavaSound libraries which can be used instead of QuickTime for MIDI communication and to add audio features. LEMu has been programmed in such a way that outputting to realtime jMusic Audio (available with 1.3 of Java and 1.1 of jMusic) is trivial. However the extension of LEMu into the jMusic Audio realm will be put on hold until most, if not all of our features have been developed further.

As it is a live performance tool, live manipulation techniques also have to be coherent and usable, with respect to all controller sources. The program has also lacked sufficient testing on different platforms, something that will be done by the end of this year.

Summary

Although happy with many of the concepts discussed in this paper, the next six months will see most of these ideas either refined or developed. To create a coherent and user-friendly program, with a commercial prospect in mind, we need to filter complex ideas into simplified user controlled parameters and develop a better understanding of programming methods and concepts. As musicians, we see that the development of LEMu is creating some new interpretations of electronic music allowing us to experiment with and realise our musical ideas

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WAVEFORM 2001

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ARTIST TALKS

Art Imitates Life's Imitation of Art

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This artist's talk describes the undertaking of a project with a working name of 'Vital Presence' at a large government funded research institution somewhere in Japan. To maintain anonymity as requested, the artist is herein referred to as *Frankenstein-san*, his collaborators as *Igor1-san, Igor2-san, Igor3-san, . . . Igor6-san* respectively. The Institute shall be likewise referred to as *The Bunker*.

Frankenstein-san was fortunate enough to have an exhibition in Paris[1] when he met the president of *The Bunker*. The President liked *Frankenstein-san's* work and arranged for him to visit *The Bunker* as visiting researcher and artist. The objective was to make an interactive virtual world where music was produced using the genetics and behaviours of artificial organisms. The *Vital Presence* project was so named because the artist wanted to make artefacts that leave the observer with the feeling of being in the presence of a living thing. Such artefacts could exist somewhere at the boundary between animate and inanimate, the world of the made and the world of the born. Exploiting the power of sound as a vital sign, a number of experimental musical environments were produced.

Igor1-san and *Igor2-san*, two student intern researchers from USA and Canada, wrote the first world on a Silicon Graphics O2 workstation. This world had some interesting features but also some serious flaws due to inexperience of the artist and programmers. The second world was made in collaboration with *Igor3-san*, a young Computer Science student from Canada, and *Igor4-san*, an alife composer and veteran MDI tinkerer from Sweden. This collaboration went much better and resulted in a surreal landscape that the user navigates with a joystick and allows intervention in the life of the creatures. Running under Windows 98, user can pick up creatures and take them to food or induce them to mate with another creature to practice a crude form of musical eugenics[2].

The third world was made as a development on the second one, but quickly became a separate project. In this project, *Frankenstein-san's* collaborators were *Igor5-san*, an energetic computer science lecturer from *Clayton's University*[3] in Australia and *Igor6-san*, a graduate student in computer science from Thailand who attended university in the USA. This world employs a more abstracted spherical depiction of the world and the creatures themselves are animated splotches of colour reminiscent of the strokes of impressionist painters. Interaction is more rarefied, involving mostly moving around and listening. It is intended that the user eventually be able to paint in patches of colour on the sphere with a drawing tablet. Creatures may be affected in unpredictable ways by these colours depending on their genetic makeup. Emphasis is more on slight perturbations of a continuous system rather than direct manipulation or control.

If time permits, *Frankenstein* will also describe other projects undertaken by him at *The Bunker*, including a mixed reality midi interface and a 3d camera tracking movement interface.

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[1] This was a presentation of an earlier alife music work produced in Australia in the late 1990s.
[2] The word *eugenics* got a lot of bad press, what with Hitler and all that, but we prefer to call it 'weeding God's garden'.
[3] Not the university's real name of course.

A Composer In Antarctica

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Keywords: Electro-acoustic Composition; Antarctica

Artist Statement

Recent technology in the field of sound has enabled the composer or sonic artist to analyse, explore, dissect and modify or alter every detail of sound in a way that often seems similar to the precise and specific cellular manipulations of a genetic engineer. Sonic artists can use this new technology to control the morphological and spectral elements of sound, thus providing an expressive continuum between the real, unaltered and recognisable sound, and extended transformations which yield more abstract gestures where source recognition is diffuse or dependent on context. Such specific control over the pitch and time domains of a sonic event has allowed the sonic artist to create new tensions and hierarchies that provide wonderful and vibrant opportunities for creative expression.

Biography

Chris Cree Brown is a senior lecturer at the University of Canterbury. His interests include conventional instrumental composition, electroacoustic and computer music, and inter-media art. His work Under Erebus is a result of his trip to Antarctica under the Artists to Antarctica programme run under the auspices of Antarctic New.

Currently, he been awarded a grant to build a 40% working model of an Aeolian Harp that has been designed for Christchurch city.

Standing wave sound sculpture: A compositional example

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Keywords: Standing waves; Beat frequencies; Csound; Composition

Abstract

This paper will discuss a standing wave sound sculpture which I created at Monash University in November last year. The term standing wave sound sculpture pertains to a sculpture using the materials of sound, which are the rapid vibrations of air particles. This differs from the more common definition of sound sculpture, which is a physical object that can create, or be used to create sound.

The standing wave sound sculpture intends to create in a space, in this case a stairwell, a sound environment that not only changes in time, but also changes in the ears of the observers as they move through the space.

Just like a traditional sculpture that changes appearance according to the viewer's relationship in space to the sculpture, the sound sculpture will also change it's sonic appearance as the observer moves in the space.

I created the sound sculpture specifically for the stairwell as it is made out of concrete which is highly reverberant and would not absorb the sound waves. The stairwell is also of a unique shape which makes the relationship and the activity of the standing waves within the space more complicated and unpredictable.

This paper will discuss the physical reasoning as to how this sound sculpture can exist, what technology and methodology I used in creating the sound sculpture including computer programs.

I will cite previous examples of this type of sound sculpture in order to show similarities and differences between my work and that of other composers.

Biography

Russell Goodwin is a Melbourne based composer working predominantly with electronic and concrete sound sources. He has created music for installations, film and theatre. Graduating with first class honours in composition from Monash University in 2000, he is currently researching the work of John Cage at Monash University.

Tritriadic Chimes: Bells in just intonation

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Artist Statement

Tritriadic Chimes is a sound installation consisting of a series of short pieces composed in just intonation. The pieces explore experimental tunings called tritriadic scales which have been developed by American tuning theorist, John Chalmers.

The motivation for *Tritriadic Chimes* stems from the idea of tuning an instrument according to its timbre. This interest originates from intonational experiments I first did in April 2000 using synthesised versions of a piece I wrote for solo carillon. *Tritriadic Chimes* is realised using Csound synthesis software which provides the ability to create bell timbres that are not found in a conventional carillon. Synthetic bells also invite other electronic means of propagating bell sounds that are not dependent on a fixed location, suggesting possibilities for future installations.

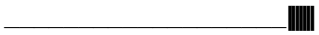
Biography

Terumi Narushima was born in Australia in 1972. She gained her undergraduate degree in music at the University of Sydney and is currently completing postgraduate studies at the Sydney Conservatorium of Music where she is studying composition with Bozidar Kos and Greg Schiemer. She has written music for koto, tuba, accordion and carillon as well as short film, and her compositions have been broadcast on Australian radio and television.

[c]ccp Netochka Nezvanova.

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detektion of onset + offset asynchrony in multikomponent komplexes.
diskriminating standard stimulus from signal stimulus
audibility as crisis and kommodification in sine.x



dze no!sz ztruktur as system degraded to an abstract totality
inspired by desire to actualize itself as it migrates into tension with other
its audibility as subject, is imbued with urgency evoked by experience \ mem[e]module. i\o
its central impulses mobilized by the objective tendencies of the crisis itself.

accelerating along 0+1 hyperbolic channel the system resists delimitation
in an obscure anti-synthesis will of force.
cognizant of its audibility the audible gains awareness of
self-internal structure, habitation.
its actualization - its loss of maschinik autonomy, incurs an internal dialectic:
retain self-alienation.
experience internal kommodities simultaneously with event, whilst posited as
nearfield patterns engaged in reflective displacement.

its impulses manifested in exterior structures
it is alienated under a regime of ownership
its becoming audible becomes a means of production.
the industrialization of audibility kommodified - the relationship between
labor of production and the event thermo-comprehends the linear coherence of
audibility as process, the objektivisation as generation of confronting objects
as alienation of the event to the process [kapitalization] and
the resolution of this alienation.

actualization alone permits comprehension of the essence of production
and the objective event: an actual becoming of entity - resultant of its
process of audibility.

as an existing event, in an effort to preserve self-experience it self-defines
self author of cell-development. if the process of audibility is
interpreted as the experiential basis of the kommodification of cell-processes
it eventuates recognition of the self at stages of production
permits the resolution of the alienation of temporal existence.
critical insight into the dialectic of alienated production and kommodification attains
practical efficacy within the system, arising out of the objective krisis complex.

dze no!sz ztruktur as one kommodified system of recursive discourse impedes
examination as a konsummate territorialized product.
internal events engaged in non-deterministik circulation, insist on non-fixity of form.
consequently the system's architecture may be generalized only as embracing [relays]
the means of production rezultant of the means of cell-production.
it is a massless object, an abstrakt maschine capable of self-acceleration,
its capacity dependent directly upon the practical appropriation of internal
events that are measured respektive to self-relations in discourse.
consequently a commodity's cell-fitness is formulated in terms of
its temporal audible space eventrelation ships [self]

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d!fferent!al d.ka! on 0+1 dezolate plane ov debr!z
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komplecx !nternl referentz
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flatl! abrogate object!v anal!s!z

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Cathedral Impulse: Towards a sonic art

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Keywords: Sound Art; Spatial Image; Aesthetics; Criticism

Abstract

This talk is a review of the processes and ideas that led to the development of the installation work entitled *Cathedral Impulse (work in progress)*. *Cathedral Impulse* is an experimental sound art installation. In this post-art era, sonic arts practice must face the challenge of re-establishing its criteria for evaluation. This may involve a reappraisal of the balance between aesthetic and intellectual conceptions of its existence. Equally, an assessment of the relations between visual and audible content in image making processes may be significant. Acoustic communication is a well understood field within the commercial media, and sound art cannot be viewed in isolation from its broader cultural context. Is it valid to continue to produce *experimental* work? Can sonic arts practice demonstrate the characteristics of a mature art form both in terms of its practice and its criticism? The installation *Cathedral Impulse* will be examined in terms of its dynamic spatialisation, its texture and spectro-morphological properties.

Biography

Ian Stevenson is a sound artist, designer and engineer. His work has appeared in theatres, broadcast media, and public spaces in Europe and Australia. He currently lives in Sydney.

The Trans Mutational Meta-Processor

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Artist Statement

This presentation will describe and demonstrate the Trans Mutational Meta-Processor (TMMP); a system recently built using Max/MSP/Nato software. This system was built to explore possible relationships between 3D animations and audio that would enhance character design in an abstract 3D world.

The system essentially consists of modular “patches” which perform audio or video signal processing.

Control signals (MIDI or internal abstracted data) change specific parameters of the audio and / or video “patches”.

The user may choose which specific parameters are controlled, and by what control source. This flexible method of control lets the user explore the synaesthetic qualities of audio visual synthesis.

The presentation will focus on technical and aesthetic concerns associated with the project, and a demonstration of the system and its final output will be made.

Biography

Using a diverse range of media, Troy Innocent has achieved international recognition for his digital media works which explore the ‘language of computers’.

During the past ten years, he combined computer animation, generative systems, multimedia design and interactivity to create virtual worlds. These worlds are a means of collecting together objects, characters and icons that are influenced by video games, electronic music, and digital aesthetics. In his work, Innocent explores the dynamic between the iconic ideal and the personal specific, the real and the simulated, and the way in which our identity is shaped by our language and communication.

Innocent is currently Senior Lecturer in the Department of Design at Monash University, Melbourne, Australia, and is producing Evolglyph, an interactive virtual world that generates iconic forms. He is also developing a major installation work entitled Artefact for exhibition later in the year, and undertaking a Master of Arts investigating a model for digital media language.

(No biography of Jeremy Yuille was available.)

WAVEFORM 2001

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STUDIO REPORTS

Studio report: Queensland University of Technology

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Abstract

he last 12 months have been a busy one for music at QUT. There have been numerous concerts and CDs created by students and staff, including Andy Arthurs' (HOD) compositions for the Queensland Theatre Company productions. Students continue to play regularly in Brisbane clubs and the alternative electronic music scene has grown rapidly with regular concerts at the new Powerhouse Centre for Live Arts and many events organised by Brisbane artists such as Andrew Kettle and Michael Norris in which staff and students of QUT regularly participate. We have been commissioned to do the web site sound design for this years Brisbane Festival of Music. The electronic music ensemble Simulation are touring to this conference and then to Melbourne for performances as part of our expanding emphasis on the integration of live performance and music technologies.

We continue to have strong demand for the undergraduate program and next year will expand to offer two new degrees in addition to the Bachelor of Music in Intermedia Arts and Sound Design. The former focussing on combining visual arts and music and the latter on IT, Multimedia and
Music. These changes are part of a wider shift in QUT structure with the introduction of the Faculty of Creative Industries of which music is a part.

The post graduate area has been growing and we have seen a four-fold increase in post graduate numbers (from a low base) over the past year. A significant contributor to that is the REV project (Real, Electronic and Virtual Instrument design, composition and performance). This postgraduate project focuses on elecronic and cyber instruments including software design and will culminate in a two week REV Festival in Brisbane around Easter 2002, in which we encourage all instrument makers to participate.

Our research activities continue to centre around instrument making and using (in the broadest sense). The jMusic environment continues to expand with added facilities for real time audio, JavaSound MIDI support, XML reading and writing, and peer to peer networking capabilities added or significantly expanded in the past year. jMusic is being used as the basis for an online music tools project in conjunction with Education Queensland to deliver computer music making facilities via the world wide web. Our new faculty structure has resulted in a new research centre being established, the Creative Industries Research and Applications Centre (CIRAC) and this centre is supporting the above research projects with staff and teaching relief this year.

We look forward to continued growth inthe post graduate area, finding ways to further work with interested parties in the Brisbane music community and beyond, and to generate new work and knowledge through our research efforts. The past twelve months have been productive, and the next year looks just as exciting for music at QUT.

Studio Report: University of Auckland

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Abstract

Does electronic music study mean aesthetic stifling, reliance on standardised models of 'university' computer music? This report will place particular emphasis on the undergraduate teaching activities: details of goals and methodology of our new introductory course, Music and Technology, and the 200 and 300 level papers in electronic music which account for most of our teaching. Around forty students each year are involved at various levels of studio composition that has included music for theatre, dance, film and video, and interactive compositions. Students are a mix of music majors and others from Film Studies, Science, Fine Arts, Architecture and Engineering.

Approaches to teaching, attitudes to aesthetic boundaries in assignment work, and software currently used will be discussed, and the sort of creative skill that is manifesting itself even within the boundaries of 'the assignment' will be illustrated with selections from simple project work (over 4 weeks and mostly dealing with sequenced FM sounds) by the following 200 level and 300 level students:

Daniel Peach, ***Army Ants***; Ken Merrick, ***Hub Mind***; Lisa Kim, ***Pitch Black***; Matthew Brennan, ***Horror Master***; SooYoung Lee, ***Resurfacing***; Amie Ridley, ***Time Lapse Desert***; David Lindsay, ***Windshine*** ; Edward Pickmere, ***The Eerie Enchantment*** ; Ruth Campbell, ***Desert***; Roderick Skipp.

Studio Report: University of Western Sydney

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Abstract

A report on activities of the Music Department, School of Contemporary Arts, and the UWS Designated Research Group in Time Based Art and Emerging Technologies.

Studio Report: Monash University

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Working With Sound

The *Working With Sound* project is currently under development in the School of Music – Conservatorium and the Centre for Electronic Media Arts at Monash University. This project involves the creation of an educational resource primarily aimed at undergraduate students studying: music, sound engineering, multimedia and computer science. Topics covered in *Working With Sound* include; basic physics of sound, sound synthesis, recording and mixing technology. The resource is designed in a self-paced learning format that includes theoretical information integrated into practical exercises. It features a suite of software applications that can generate and process sound. This enables students to explore, in a practical way, the generation and manipulation of sound by the direct application of information with experimentation. The resource will be demonstrated in the presentation

The School of Music – Conservatorium

The period from 2000 to 2001, has seen the development of several activities and initiatives that advance these areas of music at Monash.

- introduction of a Masters by research in composition and a PhD in composition.
- the establishment of the *Sonic Art Group* which is an ensemble based in the recording and electronic music studios in the School.
- *Electronic Antiphony* concert which will feature an 8 channel sound system supplied by System Sound (who provided the excellent sound system for the Sonic Residues concert series that was held in Melbourne last year. Scheduled for August 5 this year.

For general information about the composition program at Monash see:
http://www.arts.monash.edu.au/music/research.html